

Project #19: McChord CDC Back Up Power Building
J00578, J00580

Project Title: Install Backup Generator in Building J00578, McChord Child Development Center

IJO #: 53465DPA250156J

Project Scope:

Building J00578 is on the JBG Critical Asset List and therefore authorized to have an emergency backup generator. The building requires backup power to provide uninterrupted power to support sustained mission operations during emergency operations.

Building J00578 has a 600Amp 208Y/120 service.

Provide and install a new diesel-powered emergency generator providing stand-by power support to the whole building with a minimum set rating: 180kW, 208Y/120V, 3PH

The new generator shall have an integrated sub-base fuel tank incorporating double-wall Washington State environmentally compliant construction. The fuel capacity for the sub-base fuel tank shall provide a minimum 'run-time' of 48-continuous hours and incorporating 'Energy Efficient' Block Heaters and site gauge as sub-base fuel tank.

Generator controller shall be capable to interface the generator set to Enterprise Energy Data Reporting System (EEDRS) equipment using Modbus communications protocols in accordance with JBLM 26 27 13.10 30 Rev 11, Para 3.3.8. Generators must have a remote emergency stop switch to shut down the prime mover, and it shall be installed on the exterior of the generator enclosure per NEC Section 445.19(B). The new Generator should include electrical packages inside of the generator which included GFCI convenience receptacles and lights.

Provide and install a 3-way manual transfer switch, 600A, 208Y/120V to allow safe isolated load banking testing and connection of a portable backup generator tied in at the same time. (NEMA 3R)

Provide and install a new 600A, 208Y/120V ATS. The new ATS shall have isolation bypass with 4 pole configuration. (NEMA 3R)

Consider placing the new generator in the existing enclosure, in front of electrical/mechanical room. (Cultural concern)

Provide arc flash study for newly installed equipment.

The advanced metering system - Enterprise Energy Data Reporting System (EEDRS) and JACE require replacement to interconnect with the generator set. The intent of this is to provide the metering data and generator data to the local engineering staff first within the building automation system that can execute application programs to affect savings and able track

energy usage. Electricity derived from a generator is charged at a different rate than utility provided power. EEDRS is used to monitor generator output. It allows them to ensure the generator is performing properly and not producing unsatisfactory power that could cause damage to other building systems. The metering system in Building J00580 is in failing condition due to corrosion, power surges, lack of preventative maintenance, and general failures as display non functional, date and time in the meter get drifted, and clearance provided between name plate and the casing of the meter was not enough that create an unsafe system. Additionally, it can no longer be calibrated due to age. EEDRS JACE is a requirement per UFC 3-400-01 and JBLM Design Standards to be code-compliant with the installation of metering.

All work shall comply with the most current editions of NFPA 110, 70, 70E, and JBLM Design Standard



14TH ST

14TH ST

837 J00838A

838

Commissary

577

550

533

532

561

563

549

537

500

551

552

16TH ST

16TH ST

COMMISSARY DR

Shoppette

560

545

75545

555

J75

LINCOLN BLVD

WADS

852

851

853

855

841

FAIRWAY RD

Escape Zone

McChord CDC

580

578

578B

580S

580D

Air Museum

517

Burger King

510

J05500

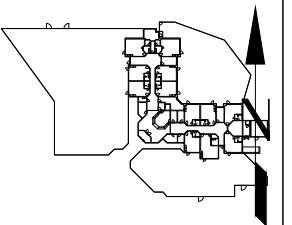
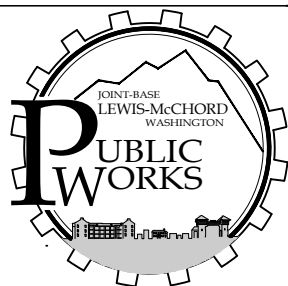
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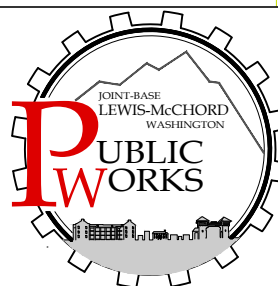
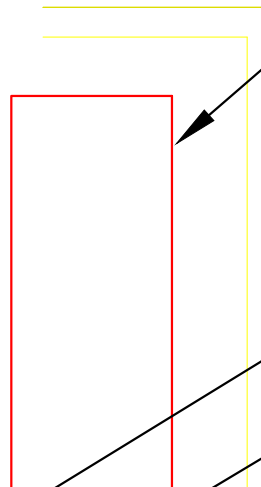
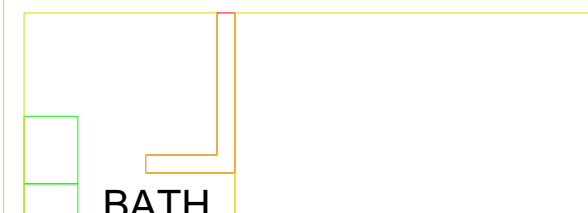
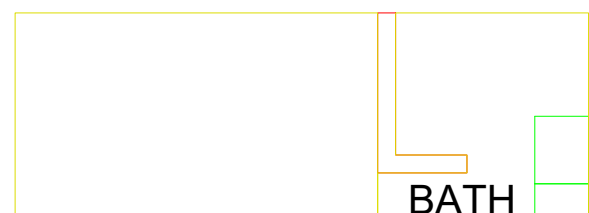
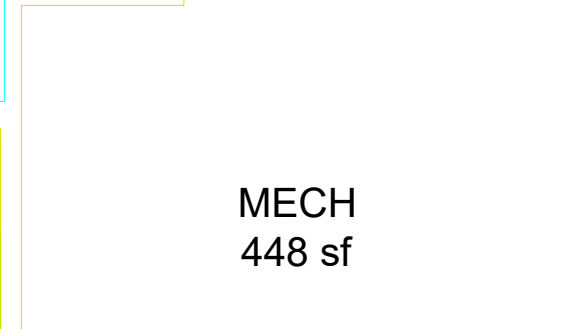
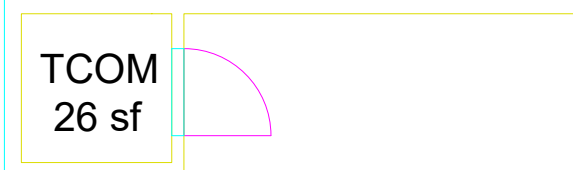
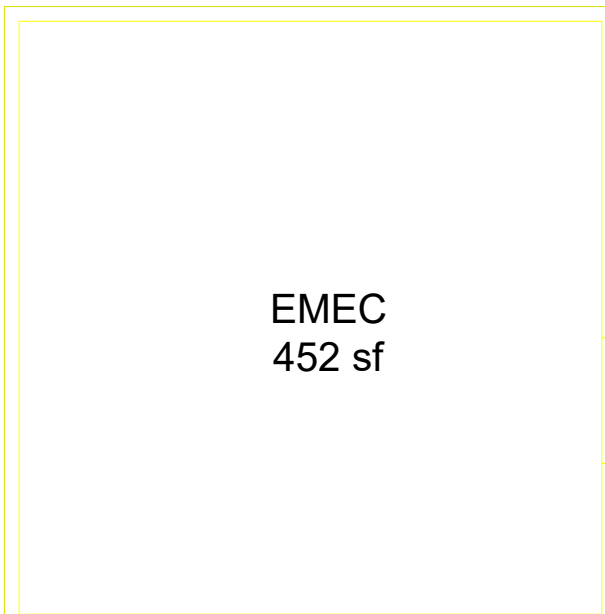
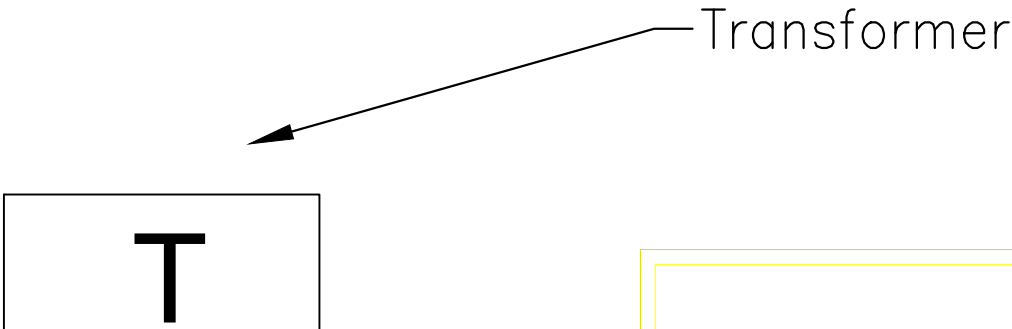
Ball Fields

516

515

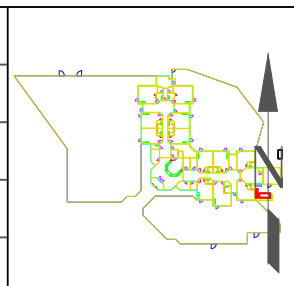
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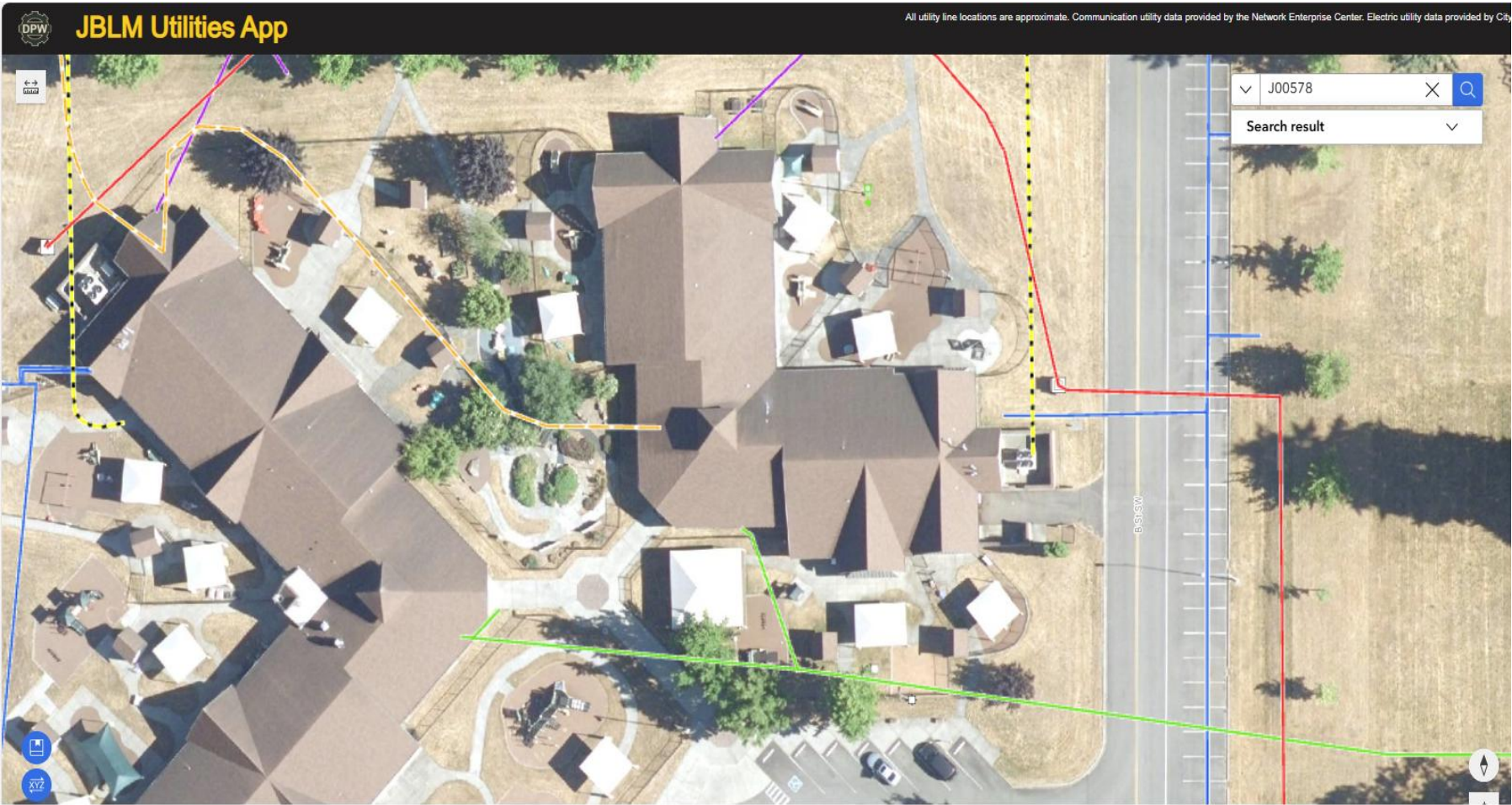


BLDG J00578
1st FLOOR
FAMILY SUPPORT (CHILD DEVELOP.)

PROPOSED GENERATOR LOCATION	
SURVEY DATE:	
NOT TO SCALE	13 January 2026 2:22:28 PM



Building J00578 Utilities



Communication



Communication - Direct Bury



Electric



Fuel



Gas



Heating or Cooling



Non-Government Utility



Stormwater



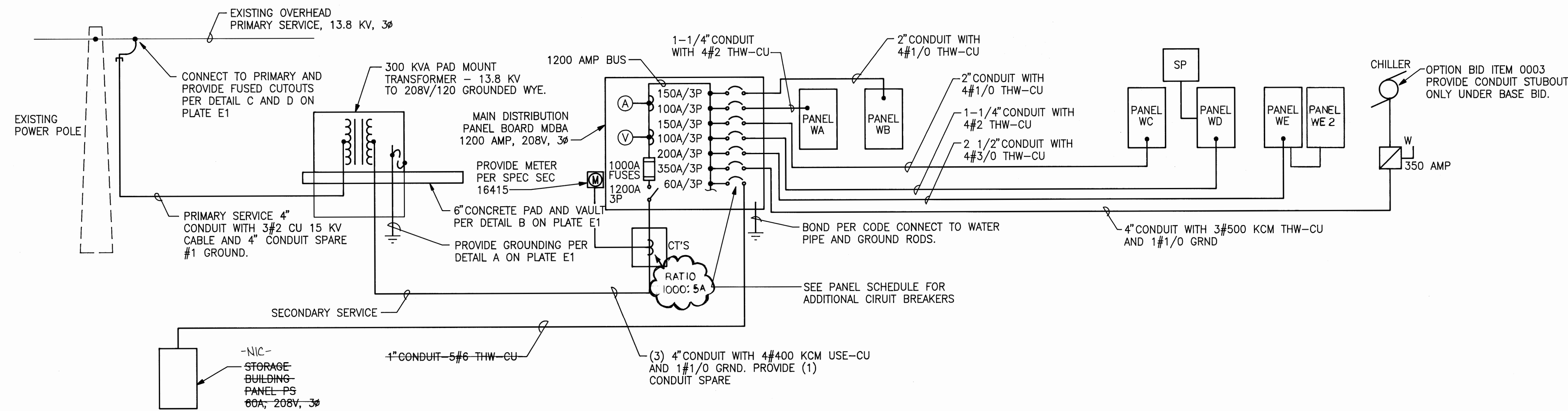
Waste Water



Water



REVISIONS				
SYMBOL	ZONE	DESCRIPTION	DATE	BY
		REVISED AS BUILT	2/16/96	TT

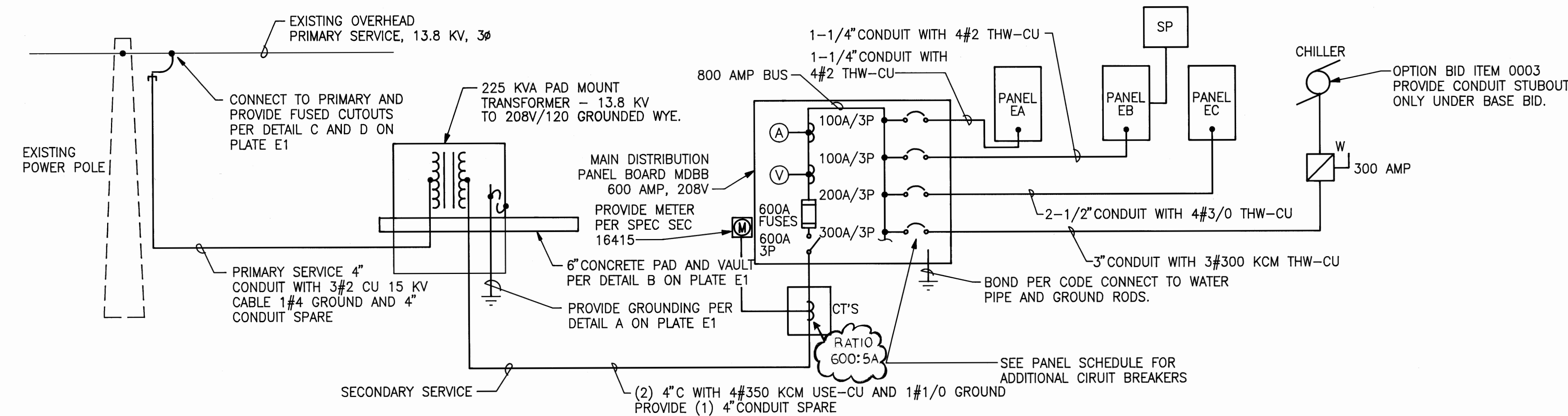


POD A – POWER RISER DIAGRAM

PANEL MDBA		3 PHASE		4 WIRE		120/208 VOLTS		6000 AMPS		MAIN FSO SW		30,000 AIC MIN.				
FLOOR		MOUNT		42 POLES		BOTTOM FEED										
LOAD	CIRCUIT	DIRECTORY	CKT NO.	CKT BRKR	A	B	C	CKT BRKR	CKT NO.	CIRCUIT	DIRECTORY	LOAD				
84680	CH-1		1	3P	•			3P	2	PANEL WE		92290				
			3			•			4							
			5	350			•	200	6							
			7	3P				3P	8							
29632	PANEL WA		9		•				10	PANEL WD		31380				
			11	100				100	12							
			13	3P		•		3P	14							
			15			•			16							
40430	PANEL WB		17	150				150	18	PANEL WC		39851				
			19	3P		•		3P	20							
			21						22							
			23	60			•	100	24							
3770	PANEL PS		25		•				26	SPARE						
			27			•			28							
			29				•		30							
			31		•				32							
			33			•			34							
			35				•		36							
			37		•				38							
			39			•			40							
			41				•		42							
		158512	TOTAL VA 322033						CURRENT		894.5 AMPS		163521			
			DEMAND VA 287002						DEMAND CURRENT		797.2 AMPS					

NOTE: PROVIDE MAIN SWITCH BOARDS WITH DEMAND METER FOR CONNECTION OF ENERGY MANAGEMENT CONTROL SYSTEM.

NOTE: SPARES TO HAVE A 200 AMP FRAME SIZE



POD B – POWER RISER DIAGRAM

PANEL MDBB		3 PHASE 4 WIRE		120/208 VOLTS		600 AMPS MAIN FSO SW		FLOOR		MOUNT		42 POLES		BOTTOM FEED		30,000 AIC MIN.	
LOAD	CIRCUIT DIRECTORY	CKT NO.	CKT BRKR	A	B	C	CKT BRKR	CKT NO.	CIRCUIT DIRECTORY	LOAD							
74800	CH-2	1	3P	•			3P	2									
		3			•			4	PANEL EB	28912							
		5	300		•		100	6									
		7	3P				3P	8									
34240	PANEL EA	9			•			10	PANEL EC	56305							
		11	100				200	12									
		13	3P		•		3P	14									
3825	CP-202 A&B 3 HP	15			•			16	CP-203 3 HP	3825							
		17	70				20	18									
		19	3P		•			20									
	SPARE	21			•			22									
		23	100					24									
		25			•			26									
		27				•		28									
		29				•		30									
		31		•				32									
		33			•			34									
		35				•		36									
		37		•				38									
		39			•			40									
		41				•		42									
112865	TOTAL VA 201907							CURRENT	560.9 AMPS								
	DEMAND VA 190519							DEMAND CURRENT	529.2 AMPS	89042							

NOTE: PROVIDE MAIN SWITCH BOARDS WITH DEMAND METER FOR CONNECTION OF ENERGY MANAGEMENT CONTROL SYSTEM.

NOTE: SPARES TO HAVE A 200 AMP FRAME SIZE

AS BUILT

**IF DRAWING SIZE IS LESS THAN 28X40
THEN THE DRAWING IS REDUCED SIZE**



THE TSANG PARTNERSHIP, INC.		U.S. ARMY ENGINEER DISTRICT, SEATTLE CORPS OF ENGINEERS SEATTLE WASHINGTON	
McCHORD CHILD DEVELOPMENT CENTER			
RISER DIAGRAMS			
McCHORD AFB		PN 3301 MC	
EDC F	DRAWING NO. B0052	FILE 226S/740-14-02	DATE 04 JUN 16
DSGN RAA	CHK MDB	SHEET 108	E11

Project Title: Install Backup Generator in Building J00580, McChord Child Development Center

IJO #: 53465DPA250275J

Project Scope:

Building J00580 is on the JBG Critical Asset List and therefore authorized to have an emergency backup generator. The building requires backup power to provide uninterrupted power to support sustained mission operations during emergency operations.

Building J00580 has a 1000Amp 208Y/120 service.

Provide and install a new diesel-powered emergency generator providing stand-by power support to the whole building with a minimum set rating: 300kW, 208Y/120V, 3PH

The new generator shall have an integrated sub-base fuel tank incorporating double-wall Washington State environmentally compliant construction. The fuel capacity for the sub-base fuel tank shall provide a minimum 'run-time' of 48-continuous hours and incorporating 'Energy Efficient' Block Heaters and site gauge as sub-base fuel tank.

Generator controller shall be capable to interface the generator set to Enterprise Energy Data Reporting System (EEDRS) equipment using Modbus communications protocols in accordance with JBLM 26 27 13.10 30 Rev 11, Para 3.3.8. Generators must have a remote emergency stop switch to shut down the prime mover, and it shall be installed on the exterior of the generator enclosure per NEC Section 445.19(B). The new Generator should include electrical packages inside of the generator which included GFCI convenience receptacles and lights.

Provide and install a 3-way manual transfer switch, 1000A, 208Y/120V to allow safe isolated load banking testing and connection of a portable backup generator tied in at the same time. (NEMA 3R)

Provide and install a new 1000A, 208Y/120V ATS. The new ATS shall have isolation bypass with 4 pole configuration. (NEMA 3R)

A brick enclosure around the new Gen/ATS/MTS to match existing brick wall and building materials should be considered in order to prevent altering the character of the space. (Cultural concern)

Provide arc flash study for newly installed equipment.

The advanced metering system - Enterprise Energy Data Reporting System (EEDRS) and JACE require replacement to interconnect with the generator set. The intent of this is to provide the metering data and generator data to the local engineering staff first within the building

automation system that can execute application programs to affect savings and able track energy usage. Electricity derived from a generator is charged at a different rate than utility provided power. EEDRS is used to monitor generator output. It allows them to ensure the generator is performing properly and not producing unsatisfactory power that could cause damage to other building systems. The metering system in Building J00580 is in failing condition due to corrosion, power surges, lack of preventative maintenance, and general failures as display non functional, date and time in the meter get drifted, and clearance provided between name plate and the casing of the meter was not enough that create an unsafe system. Additionally, it can no longer be calibrated due to age. EEDRS JACE is a requirement per UFC 3-400-01 and JBLM Design Standards to be code-compliant with the installation of metering.

All work shall comply with the most current editions of NFPA 110, 70, 70E, and JBLM Design Standard



837 J00838A

838

Commissary

577

COMMISSARY DR

16TH ST

Shoppette

560

545

75545

16TH ST

500

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552

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556

555

J75555

J71588

855

McChord
CDC



580

578

578B

580S

580D

Air
Museum

517

B ST SW

J05500

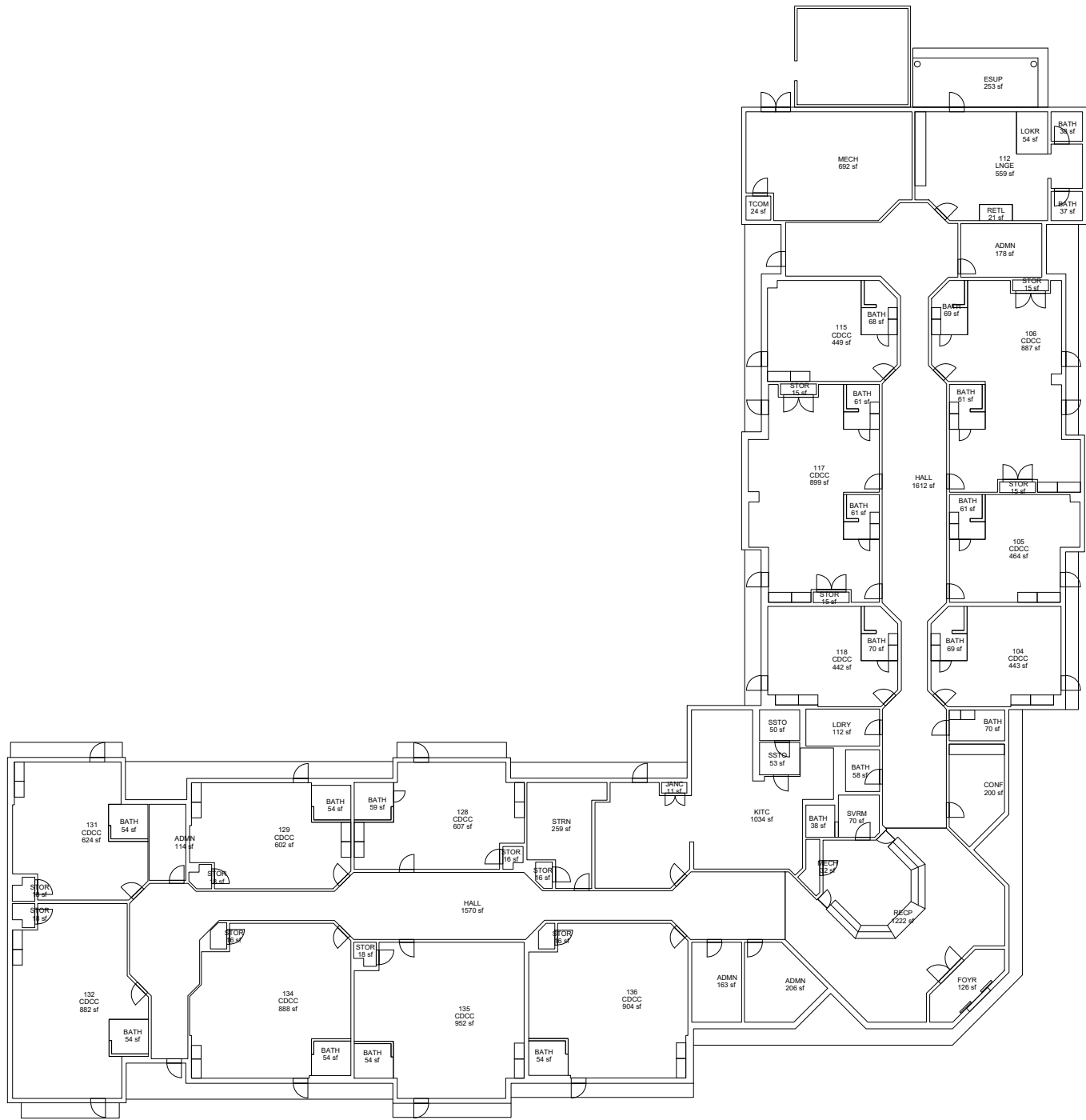
510

Burger
King

Ball
Fields

841

FAIRWAY RD



BLDG J00580

1st FLOOR

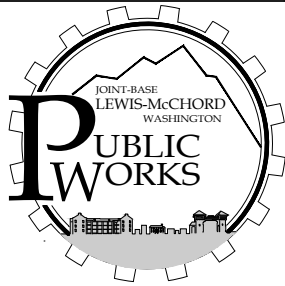
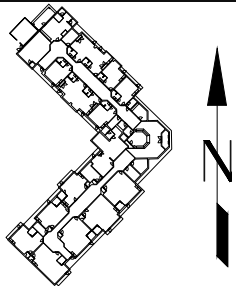
FAMILY SUP (CHILD DEVELOPMENT)

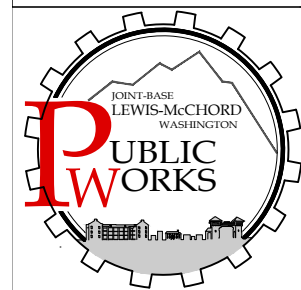
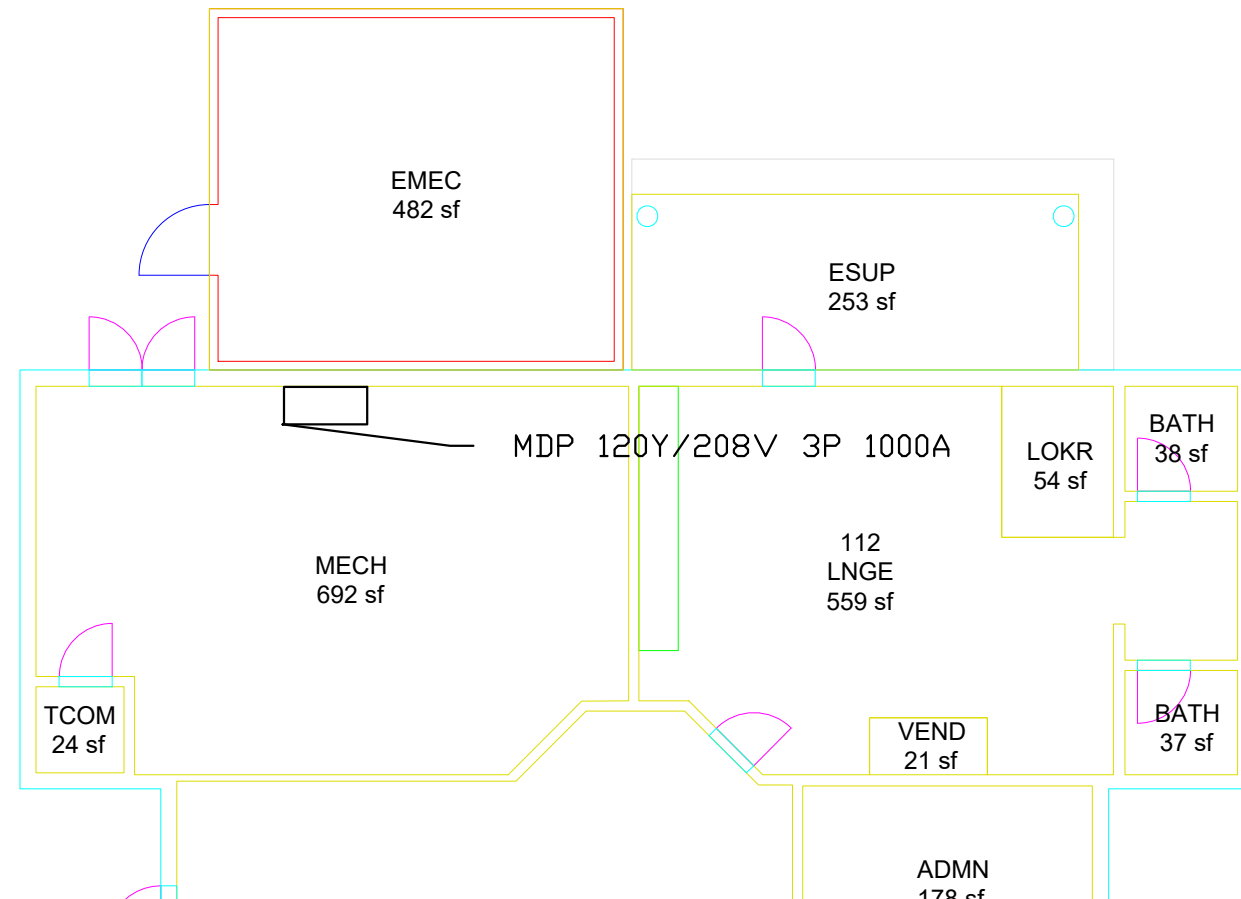
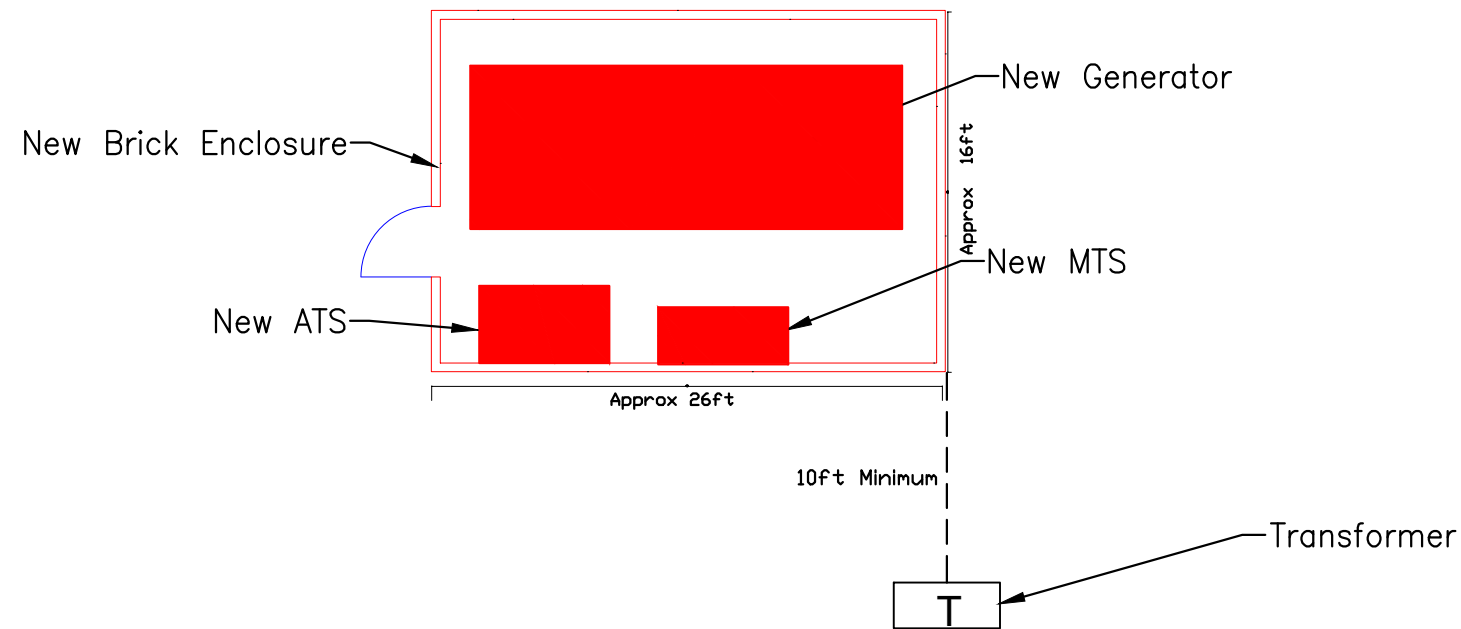
FLOOR PLAN

SURVEY DATE: 7 APRIL 2023

NOT TO SCALE

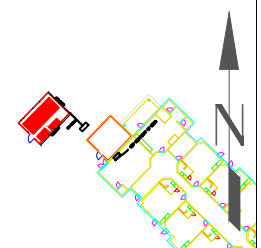
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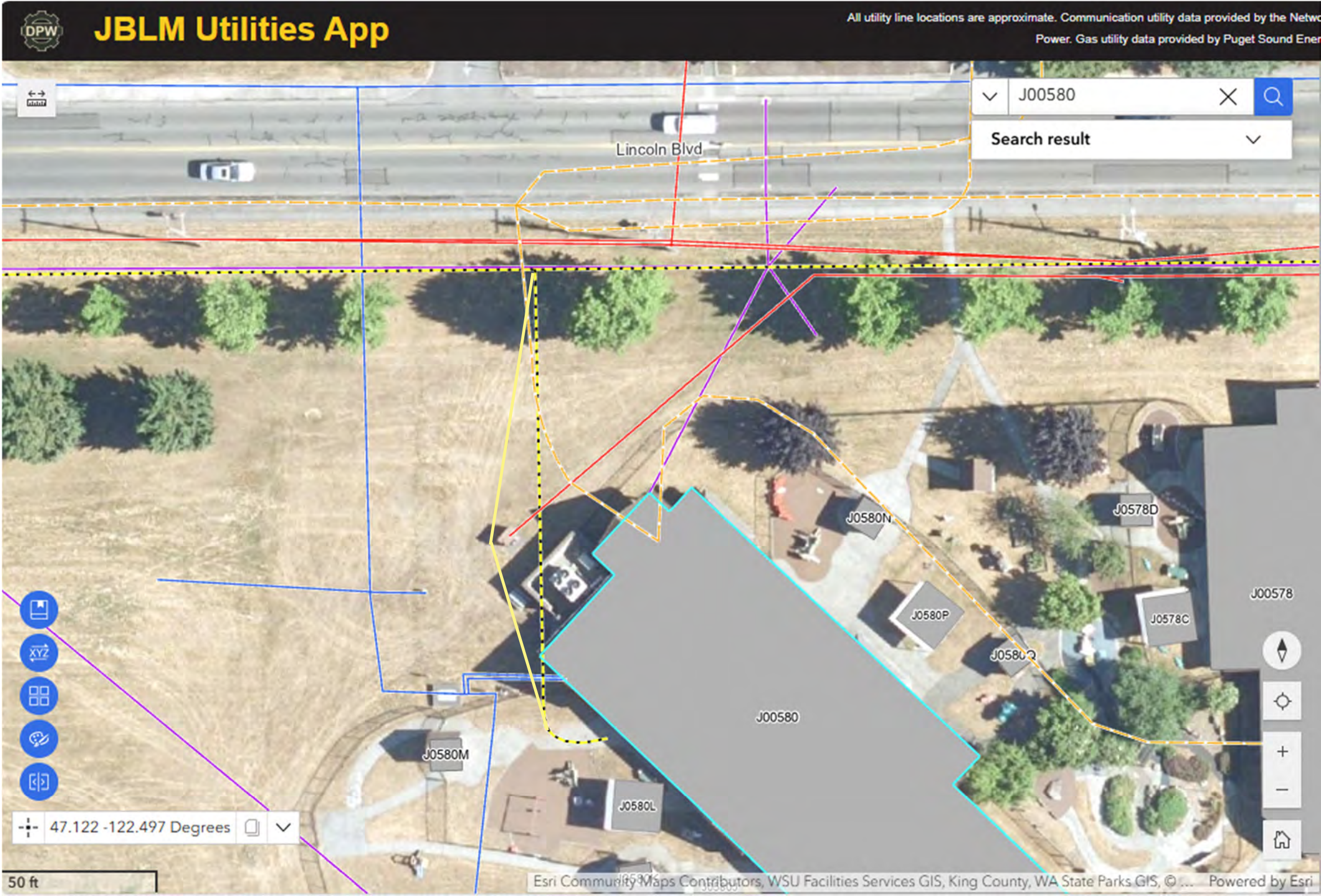


BLDG J00580
1st FLOOR
FAMILY SUPPORT (CHILD DEVELOP.)

PROPOSED GENERATOR LOCATION	
SURVEY DATE: 20 SEPT 2025	
NOT TO SCALE	25 September 2025 1:00:36 PM

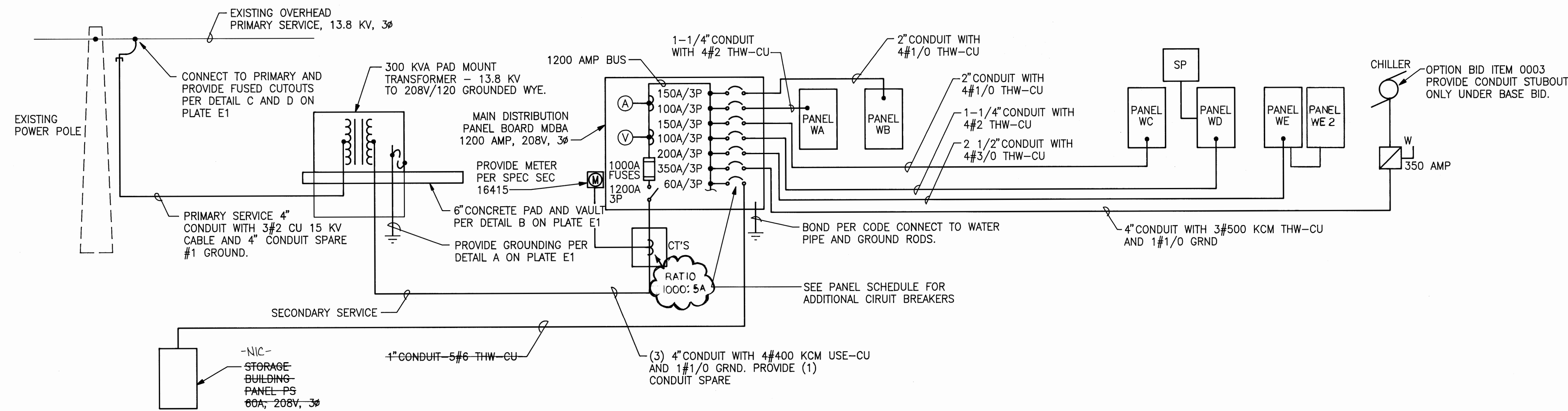


Building J00580 Utilities



- Communication
- Communication - Direct Bury
- Electric
- Fuel
- Gas
- Heating or Cooling
- Non-Government Utility
- Stormwater
- Waste Water
- Water

REVISIONS				
SYMBOL	ZONE	DESCRIPTION	DATE	BY
		REVISED AS BUILT	2/16/96	TT



POD A – POWER RISER DIAGRAM

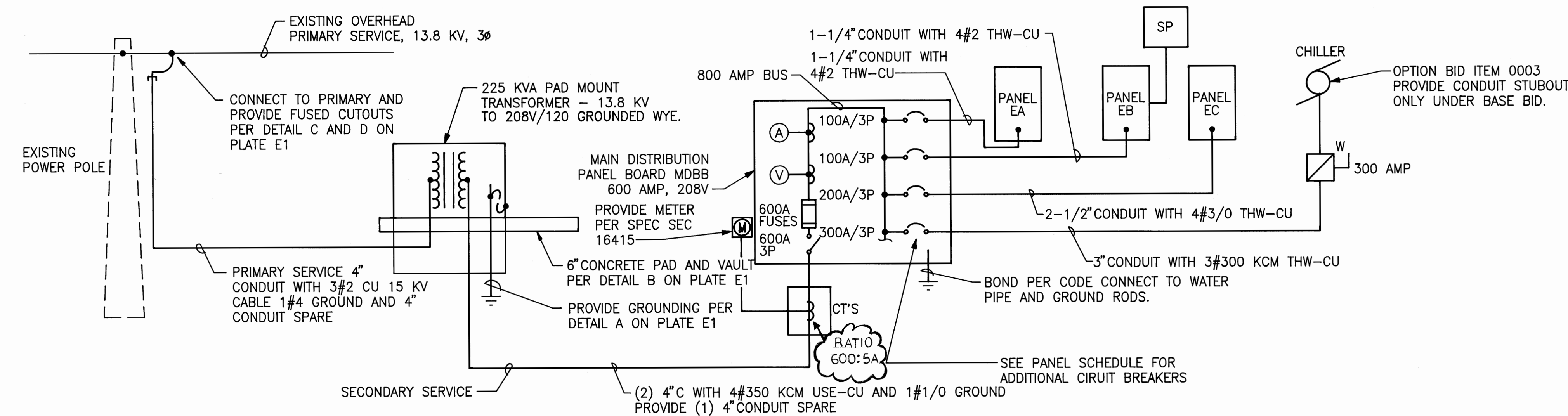
NOT TO SCALE

Building J00580

PANEL MDBA		3 PHASE		4 WIRE		120/208 VOLTS		6000 AMPS		MAIN FSO SW		30,000 AIC MIN.				
FLOOR		MOUNT		42 POLES		BOTTOM FEED										
LOAD	CIRCUIT	DIRECTORY	CKT NO.	CKT BRKR	A	B	C	CKT BRKR	CKT NO.	CIRCUIT	DIRECTORY	LOAD				
84680	CH-1		1	3P	•			3P	2	PANEL WE		92290				
			3			•			4							
			5	350			•	200	6							
			7	3P				3P	8							
29632	PANEL WA		9		•				10	PANEL WD		31380				
			11	100				100	12							
			13	3P		•		3P	14							
			15			•			16							
40430	PANEL WB		17	150				150	18	PANEL WC		39851				
			19	3P		•		3P	20							
			21						22							
			23	60			•	100	24							
3770	PANEL PS		25		•				26	SPARE						
			27			•			28							
			29				•		30							
			31		•				32							
			33			•			34							
			35				•		36							
			37		•				38							
			39			•			40							
			41				•		42							
		158512	TOTAL VA 322033						CURRENT		894.5 AMPS					
			DEMAND VA 287002						DEMAND CURRENT		797.2 AMPS					

NOTE: PROVIDE MAIN SWITCH BOARDS WITH DEMAND METER FOR CONNECTION OF ENERGY MANAGEMENT CONTROL SYSTEM.

NOTE: SPARES TO HAVE A 200 AMP FRAME SIZE



POD B – POWER RISER DIAGRAM Building J00578

PANEL MDBB		3 PHASE 4 WIRE		120/208 VOLTS		600 AMPS MAIN FSO SW		30,000 AIC MIN.		
FLOOR		MOUNT		42 POLES		BOTTOM FEED				
LOAD	CIRCUIT DIRECTORY	CKT NO.	CKT BRKR	A	B	C	CKT BRKR	CKT NO.	CIRCUIT DIRECTORY	LOAD
74800	CH-2	1	3P	•			3P	2		
		3			•			4	PANEL EB	28912
		5	300		•		100	6		
		7	3P				3P	8		
34240	PANEL EA	9			•			10	PANEL EC	56305
		11	100				200	12		
		13	3P				3P	14		
3825	CP-202 A&B 3 HP	15			•			16	CP-203 3 HP	3825
		17	70				20	18		
		19	3P	•				20		
	SPARE	21			•			22		
		23	100					24		
		25			•			26		
		27						28		
		29				•		30		
		31			•			32		
		33				•		34		
		35				•		36		
		37			•			38		
		39				•		40		
		41				•		42		
112865	TOTAL VA 201907		CURRENT				560.9 AMPS		89042	
	DEMAND VA 190519		DEMAND CURRENT				529.2 AMPS			

NOTE: PROVIDE MAIN SWITCH BOARDS WITH DEMAND METER FOR CONNECTION OF ENERGY MANAGEMENT CONTROL SYSTEM.

NOTE: SPARES TO HAVE A 200 AMP FRAME SIZE

AS BUILT

**IF DRAWING SIZE IS LESS THAN 28X40
THEN THE DRAWING IS REDUCED SIZE**



THE TSANG PARTNERSHIP, INC.		U.S. ARMY ENGINEER DISTRICT, SEATTLE CORPS OF ENGINEERS SEATTLE WASHINGTON		
McCHORD CHILD DEVELOPMENT CENTER RISER DIAGRAMS				
McCHORD AFB		PN 33011 MC		WASHINGTON
EIR F	DRAWING NO. <i>B0052</i>	FILE 2269/740-14-02	DATE 94 JUN 16	PLAN E11
DESIGN RAA	CHK MDB	SHEET 108		

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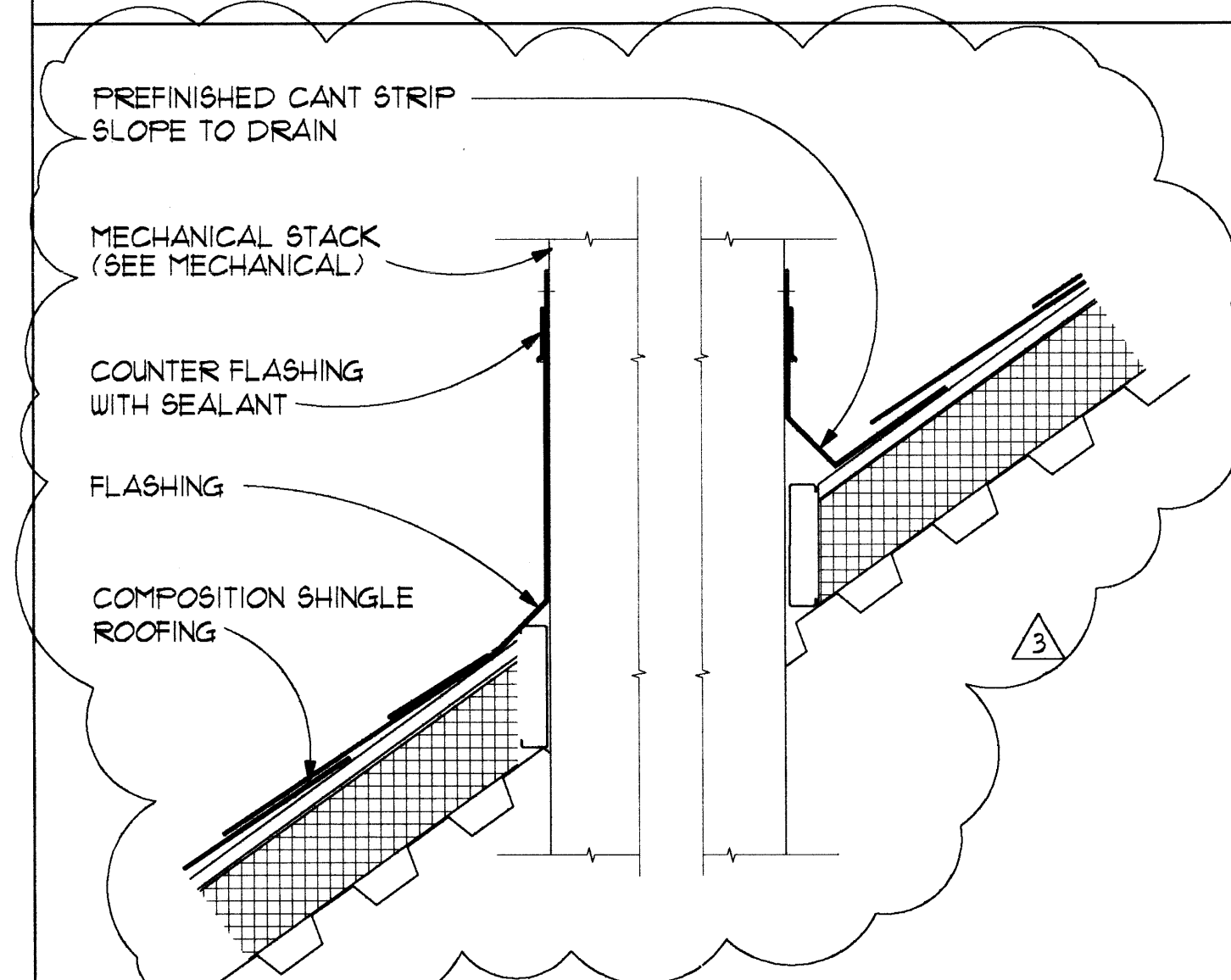
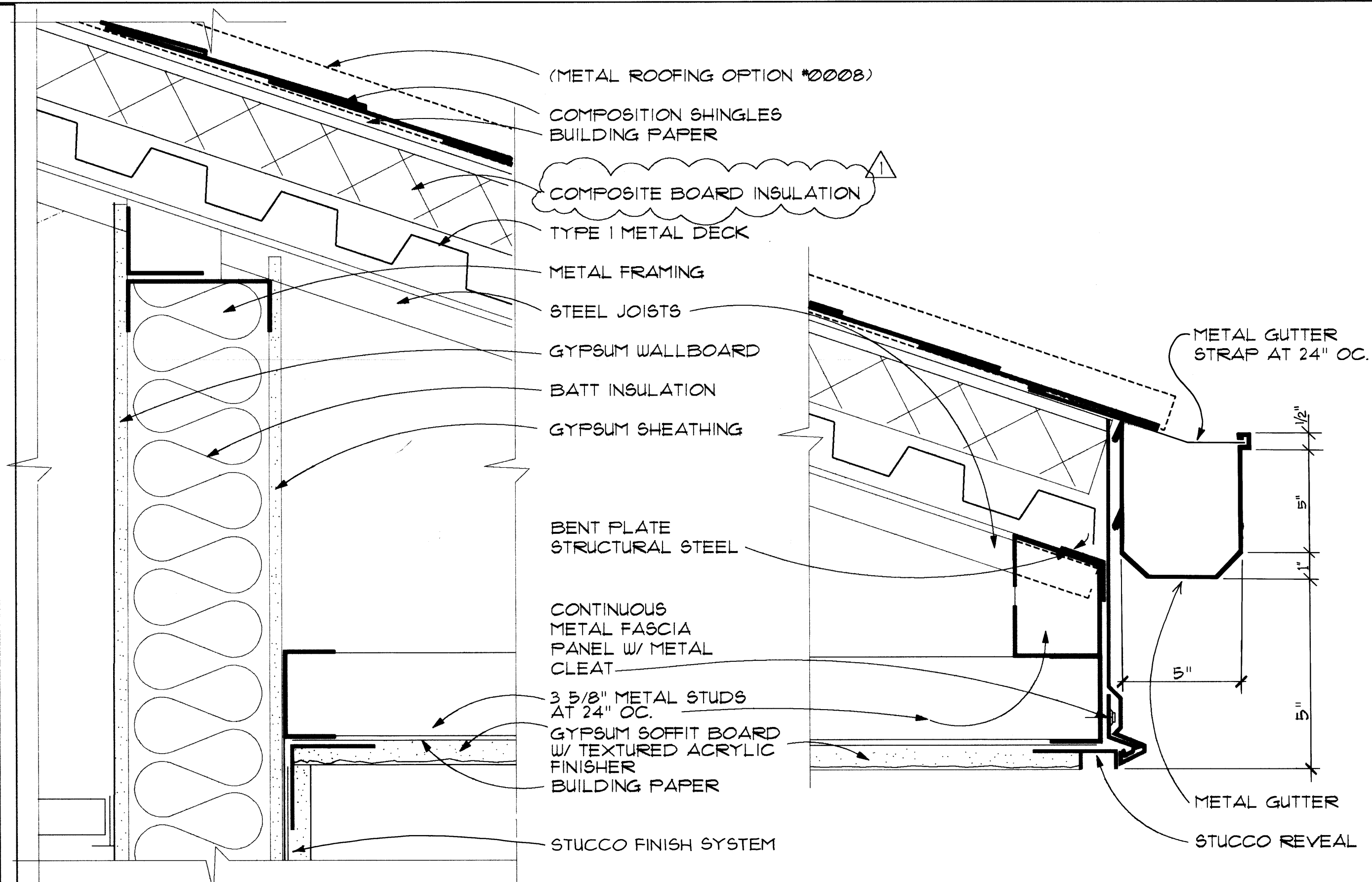
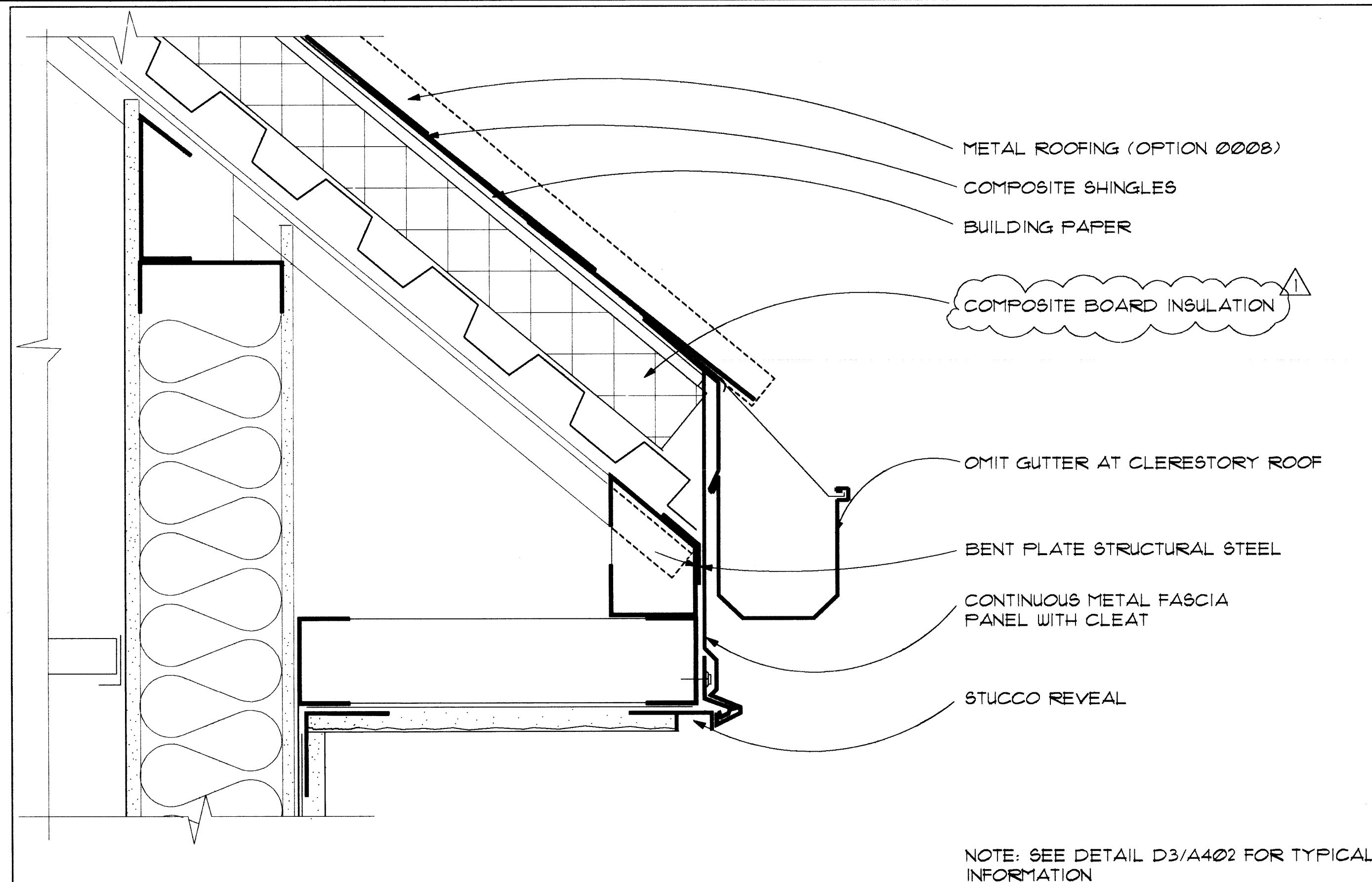
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3

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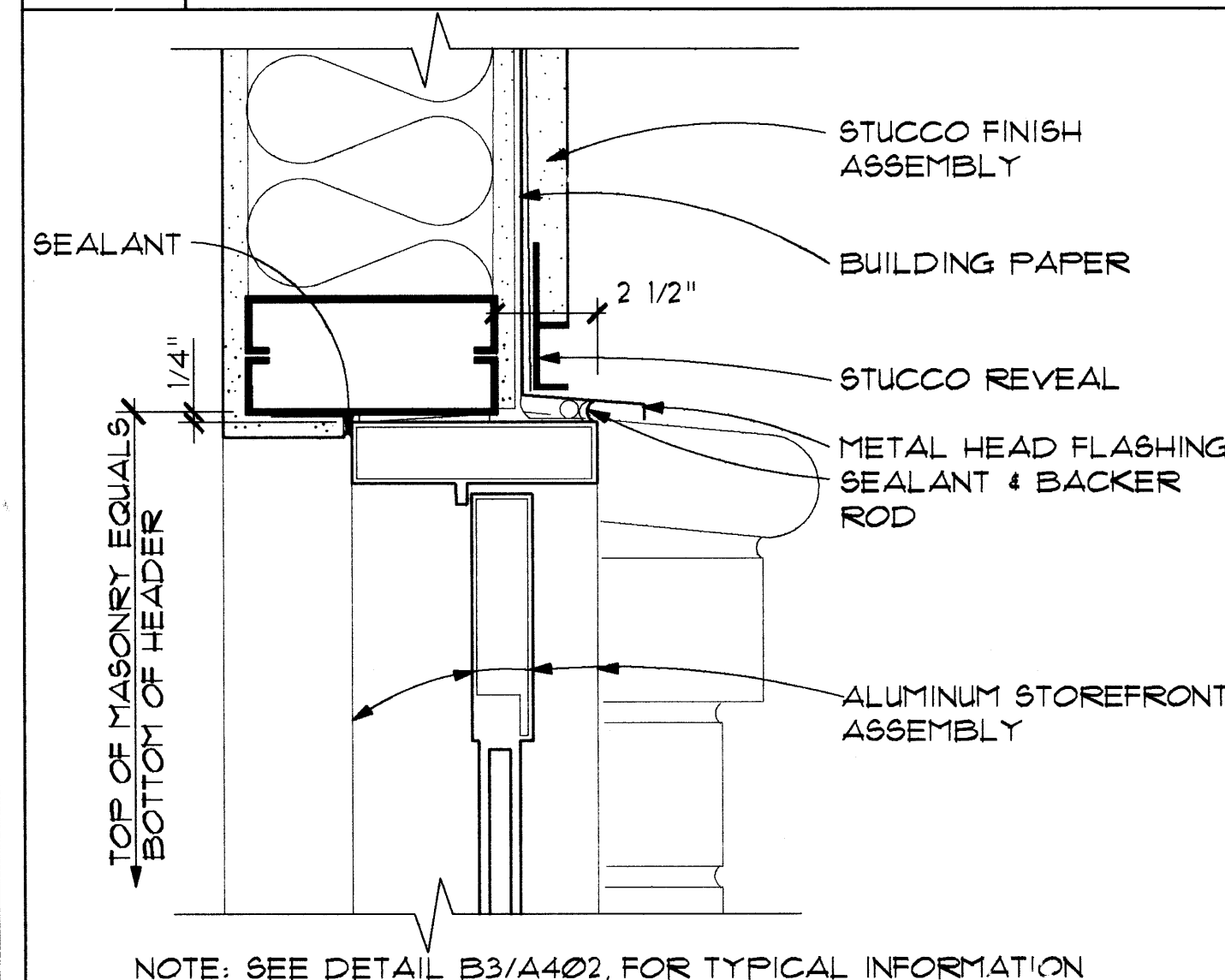
REVISIONS

SYMBOL	ZONE	DESCRIPTION	DATE	BY
△	D3/D5	CHANGE WORDING TO COMPLY WITH SPECIFICATIONS	94 JULY 5	RLR
△	B4/A4	ADDED STEEL STUD BLOCKING	94 JULY 5	RLR
△	C1	DETAIL REVISED	94 JULY 5	RLR
	SHEET	REVISED AS-BUILT	96 FEB 19	



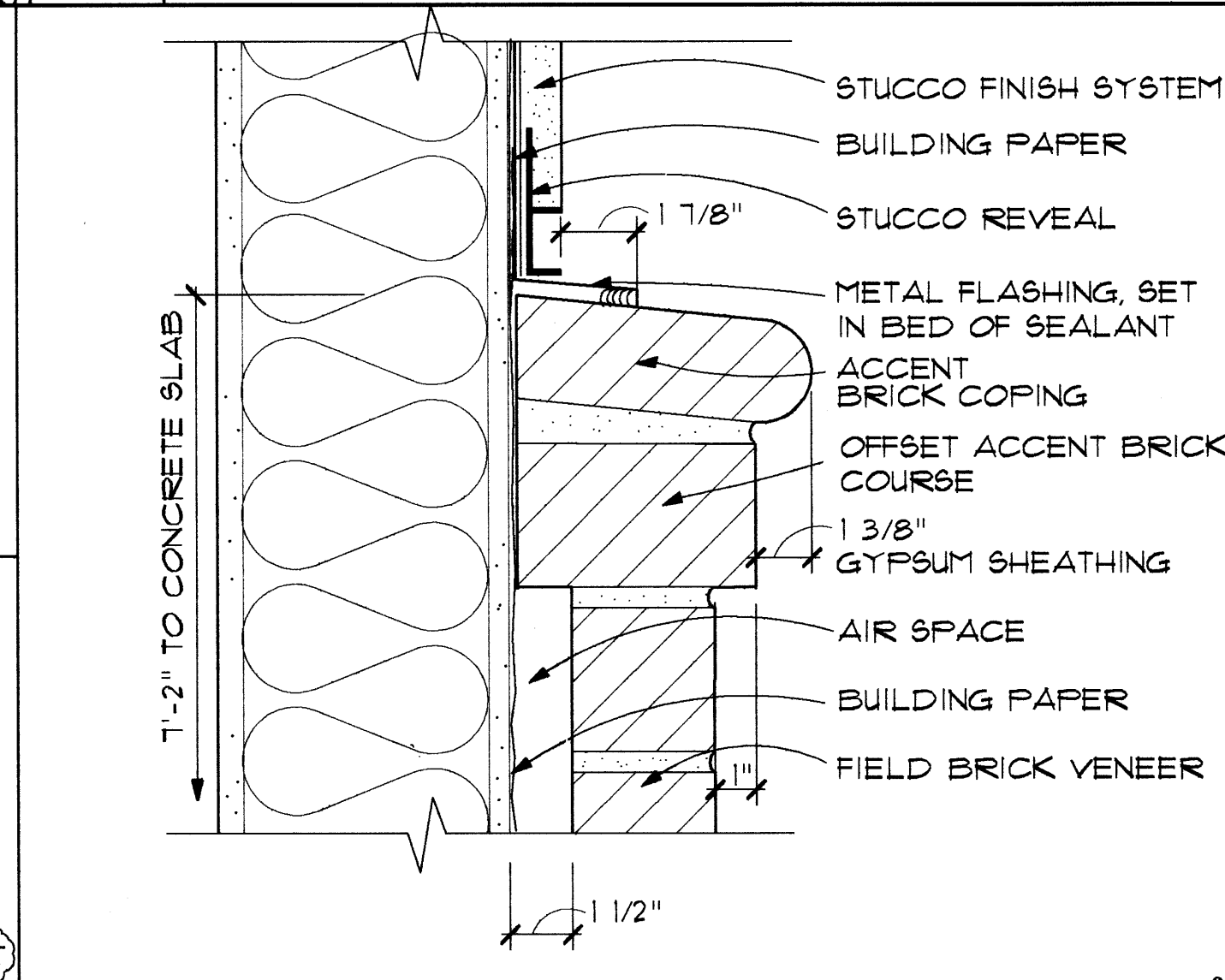
D5 SOFFIT AT GABLE

SCALE: 3"=1'-0"



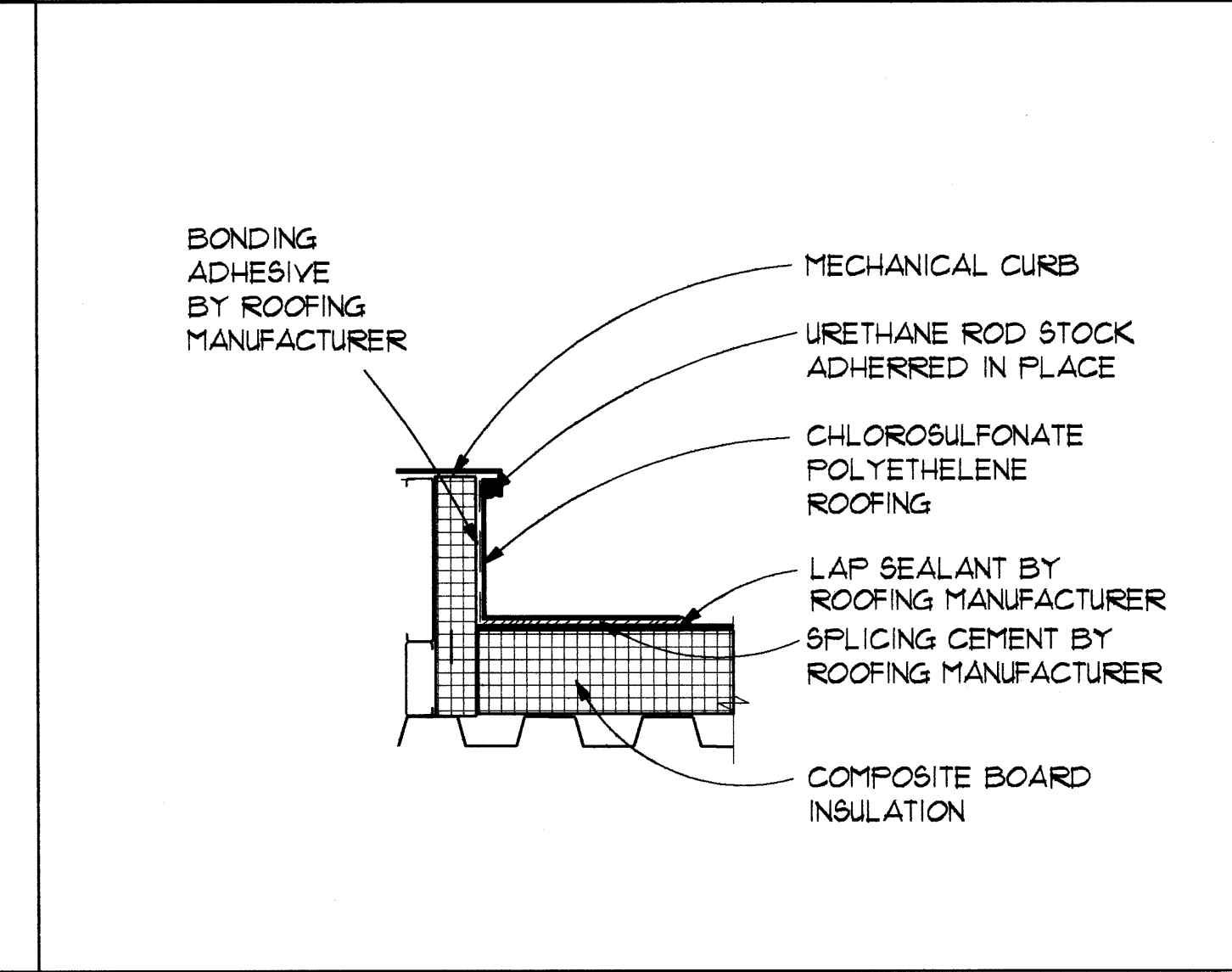
D3 TYPICAL EAVE AND SOFFIT

SCALE: 3"=1'-0"



C1 CURB AT SLOPED ROOF

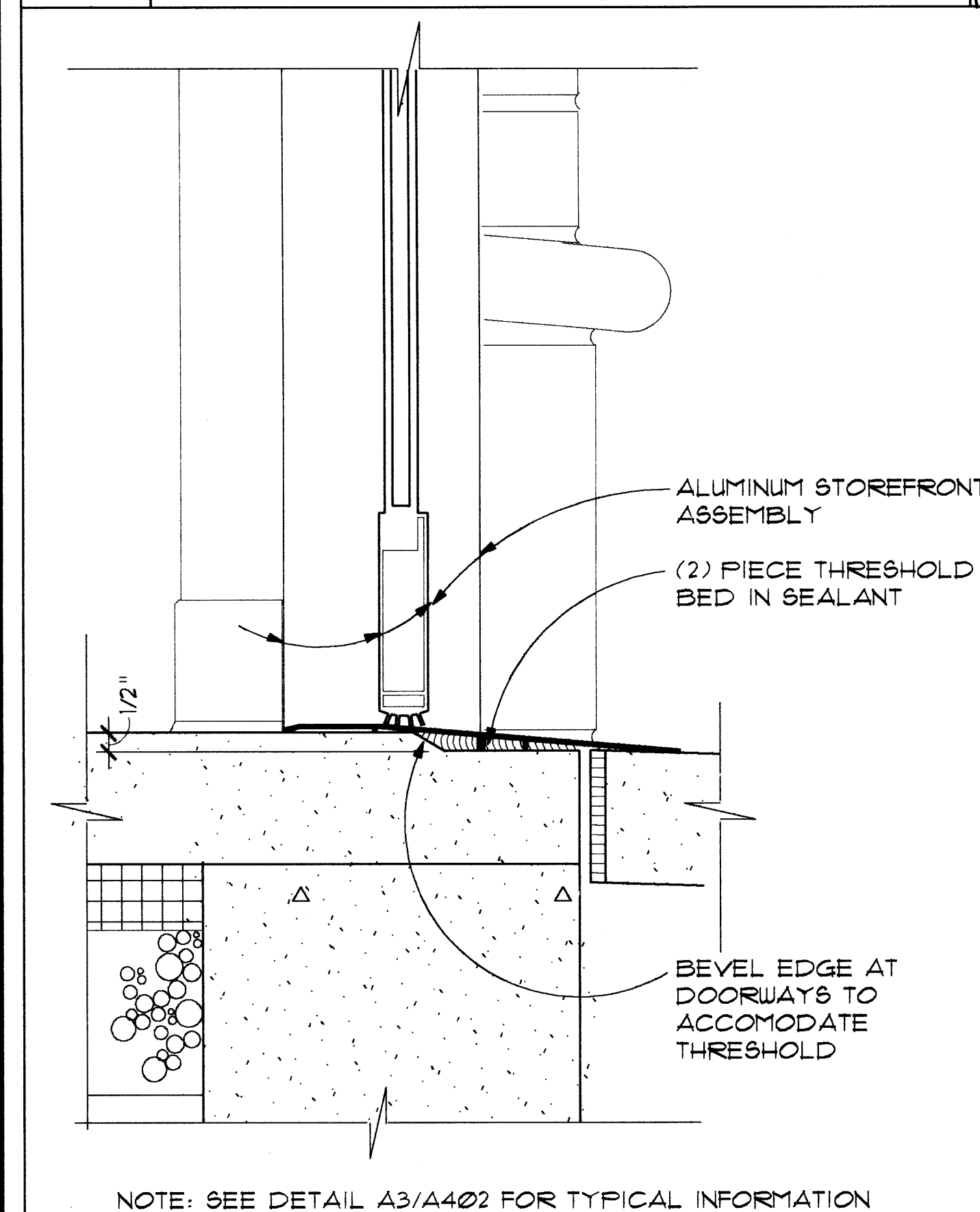
SCALE: 1 1/2"=1'-0"



B5 DOOR HEAD AT BRICK

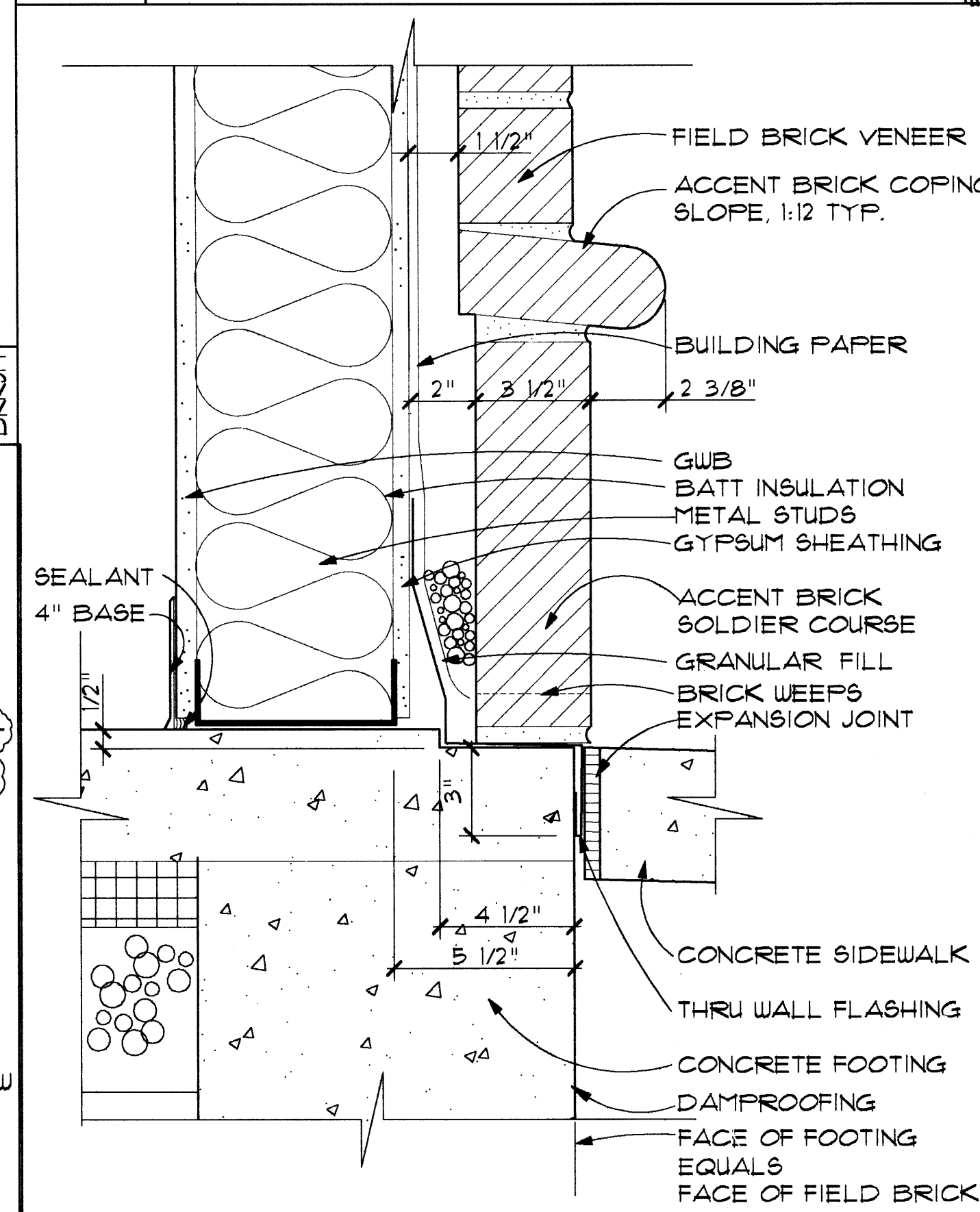
SCALE: 3"=1'-0"

NOTE: STOREFRONT WINDOW HEAD, SIMILAR



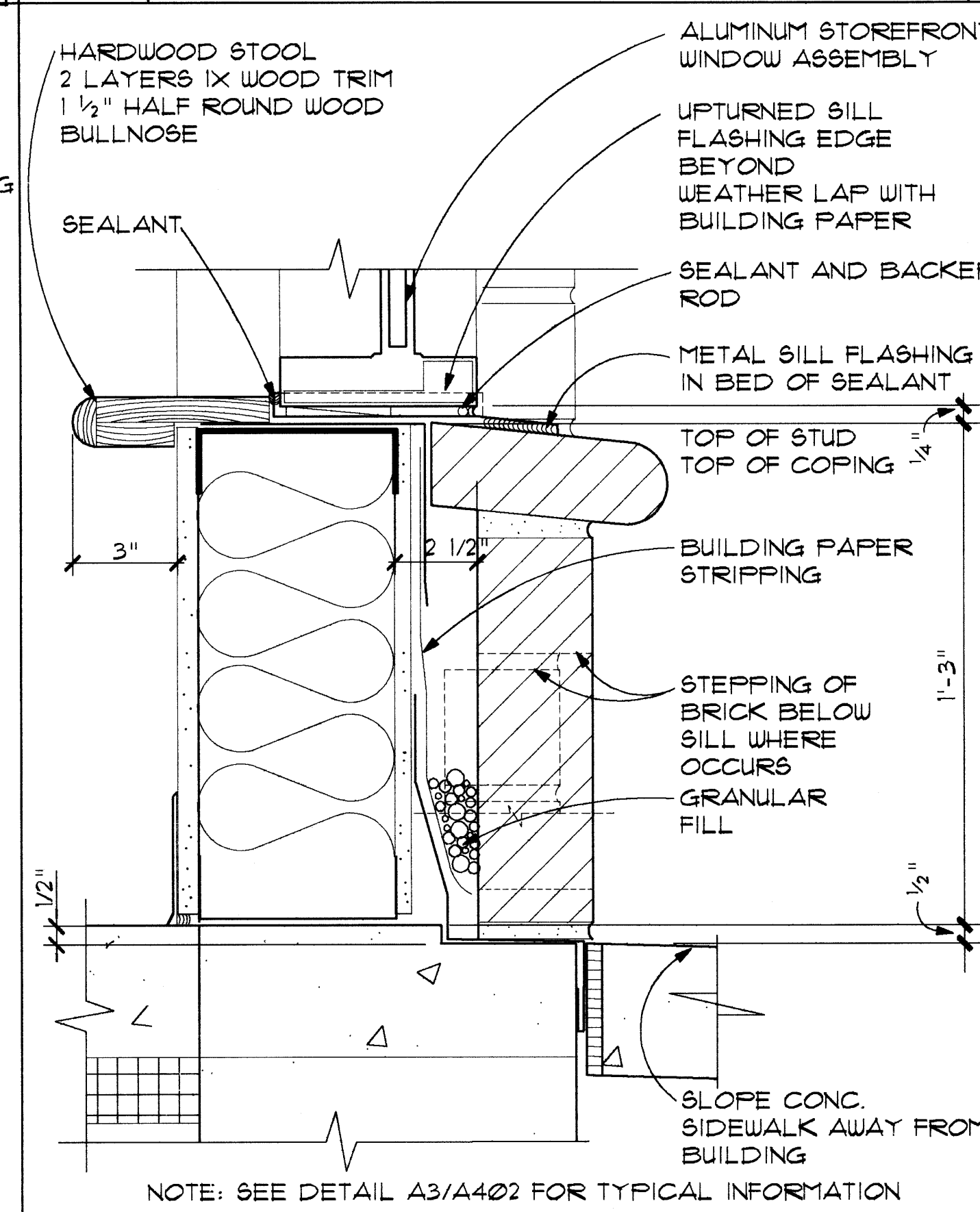
B3 BRICK COPING TO STUCCO

SCALE: 3"=1'-0"



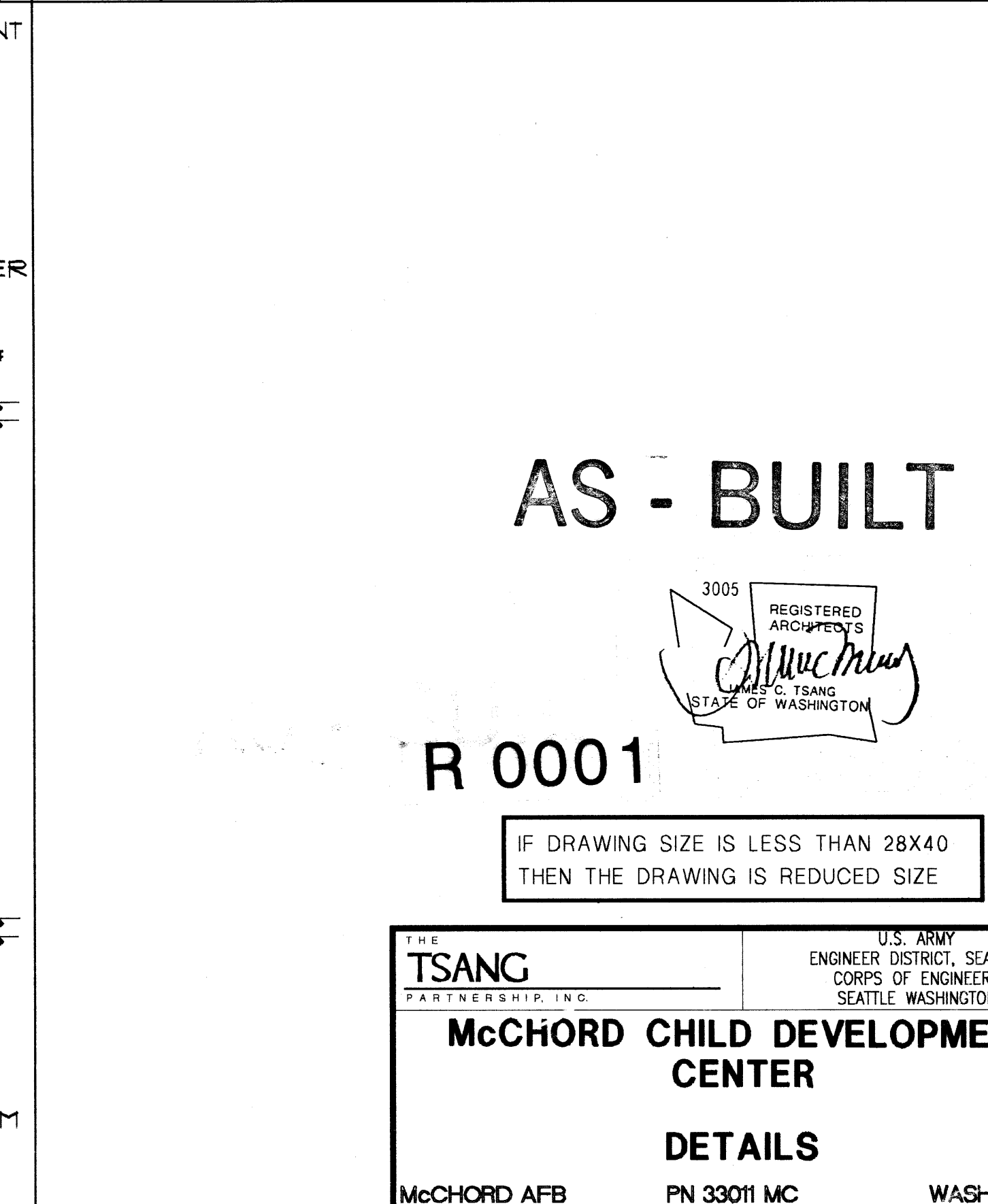
B2 WINDOW JAMB AT BRICK

SCALE: 3"=1'-0"



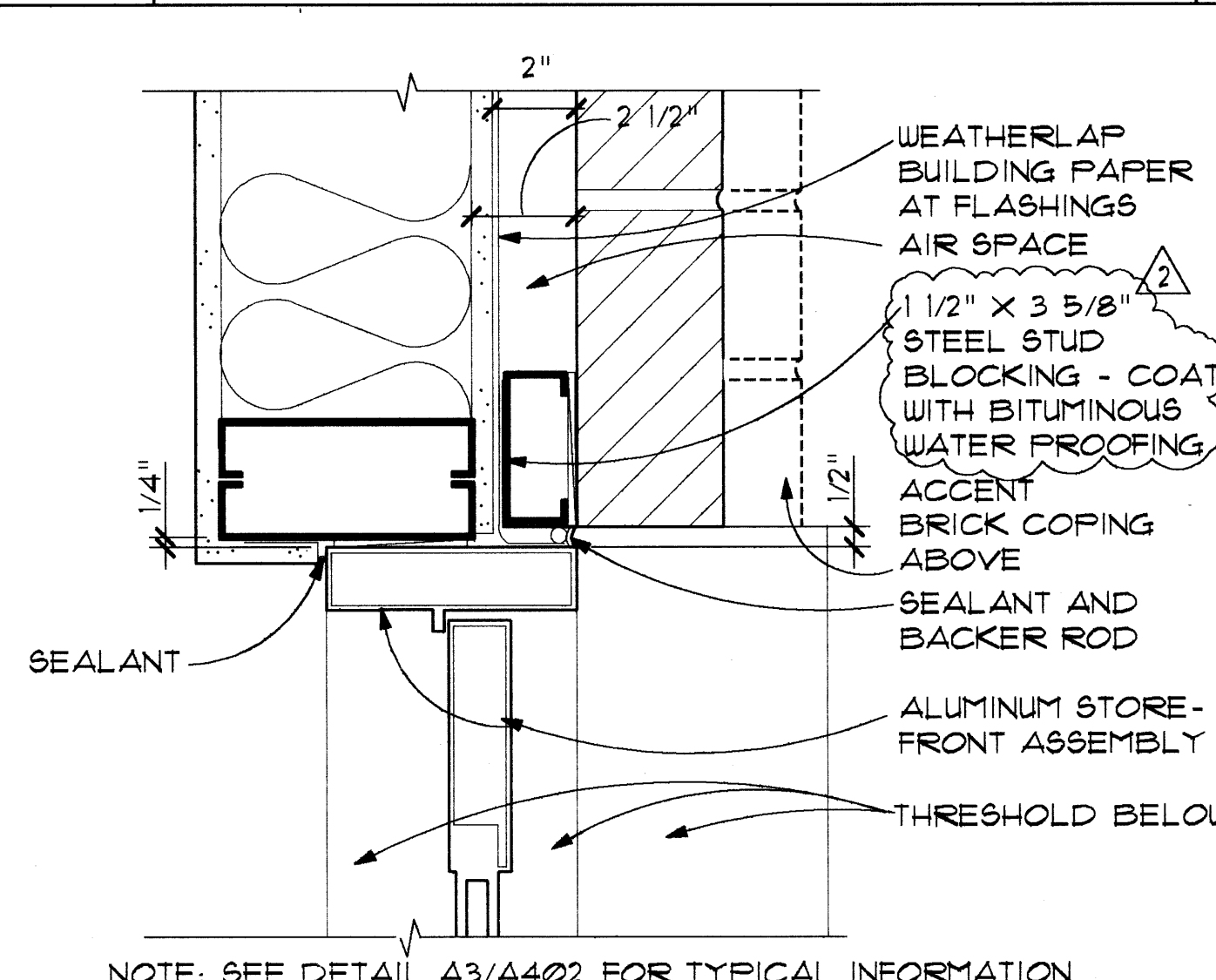
B1 MECHANICAL CURB FLASHING

SCALE: 1 1/2"=1'-0"



B4 DOOR JAMB AT BRICK FIELD

SCALE: 3"=1'-0"



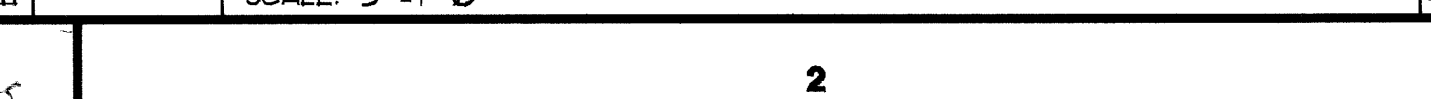
A3 BRICK SILL/COPING @ CONC. SLAB

SCALE: 3"=1'-0"



A2 WINDOW SILL AT BRICK COPING

SCALE: 3"=1'-0"



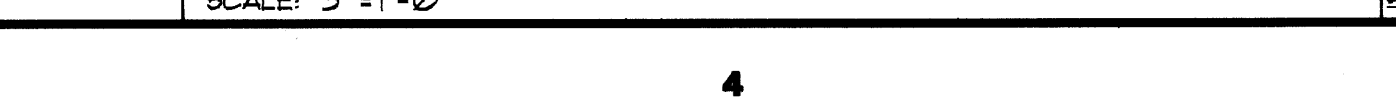
A5 DOOR THRESHOLD

SCALE: 3"=1'-0"



A4 DOOR JAMB AT BRICK SOLDIER

SCALE: 3"=1'-0"



AS - BUILT

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IF DRAWING SIZE IS LESS THAN 28X40 THEN THE DRAWING IS REDUCED SIZE

TSANG ARCHITECTS, INC.

McCHORD CHILD DEVELOPMENT CENTER

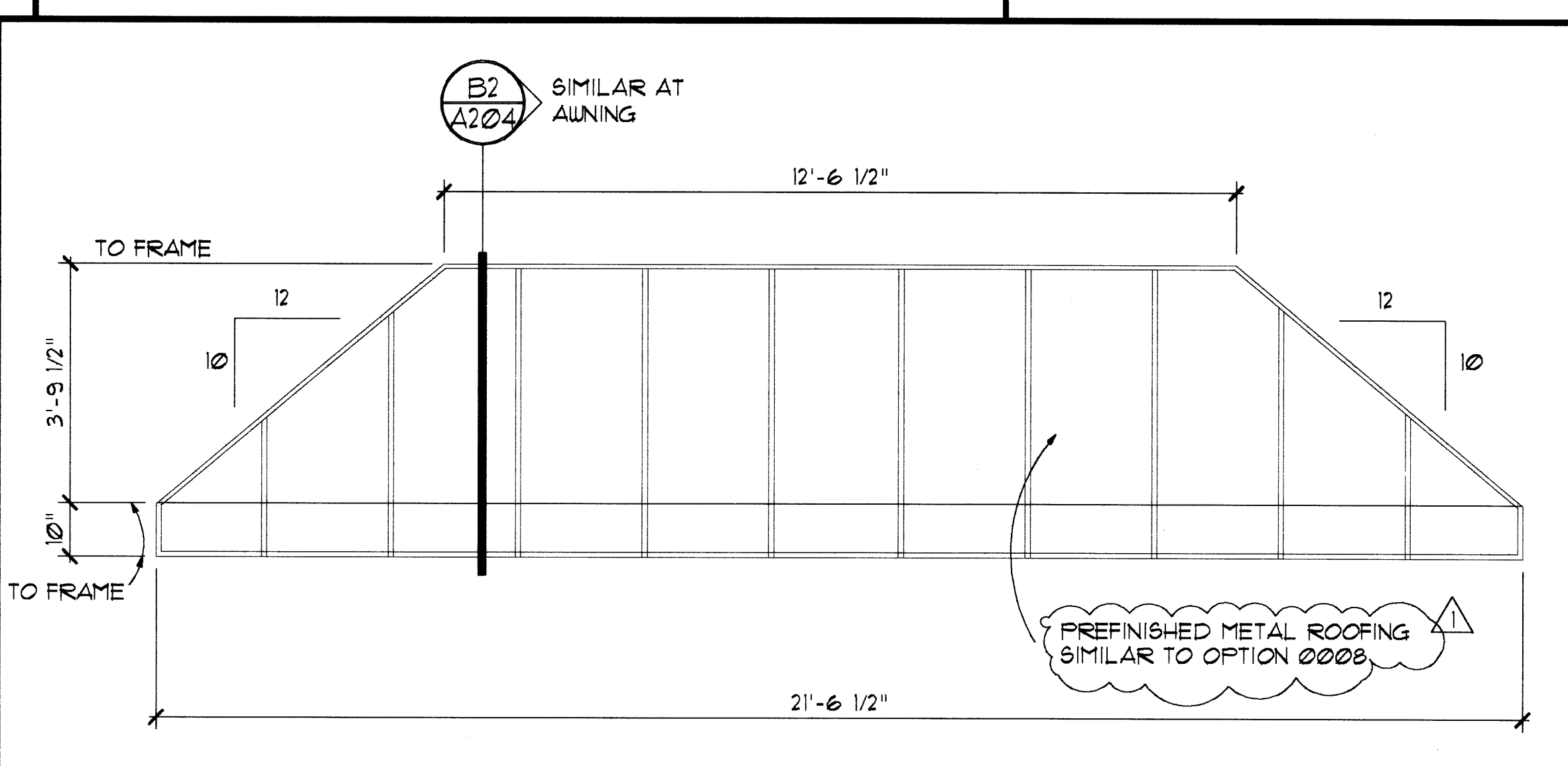
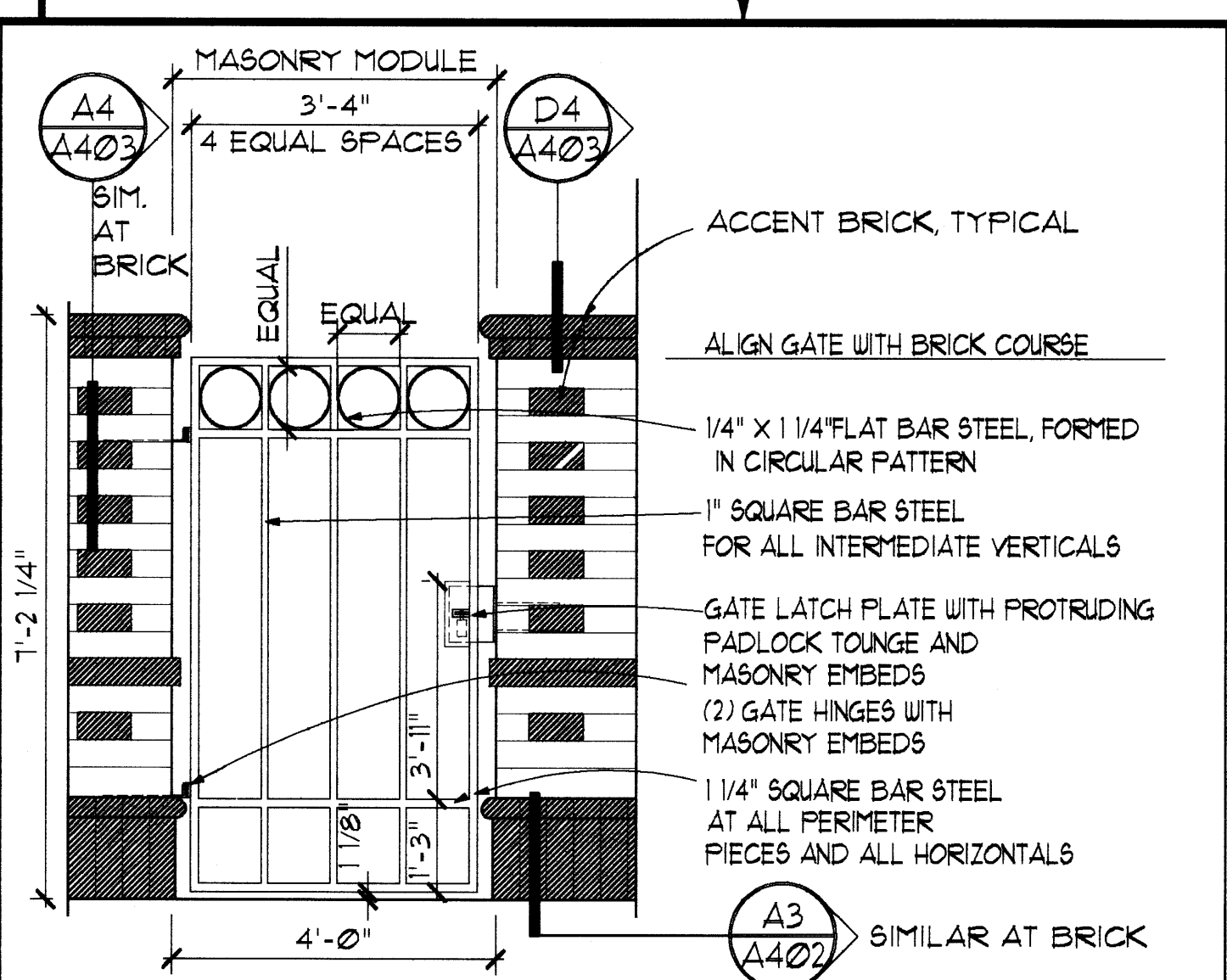
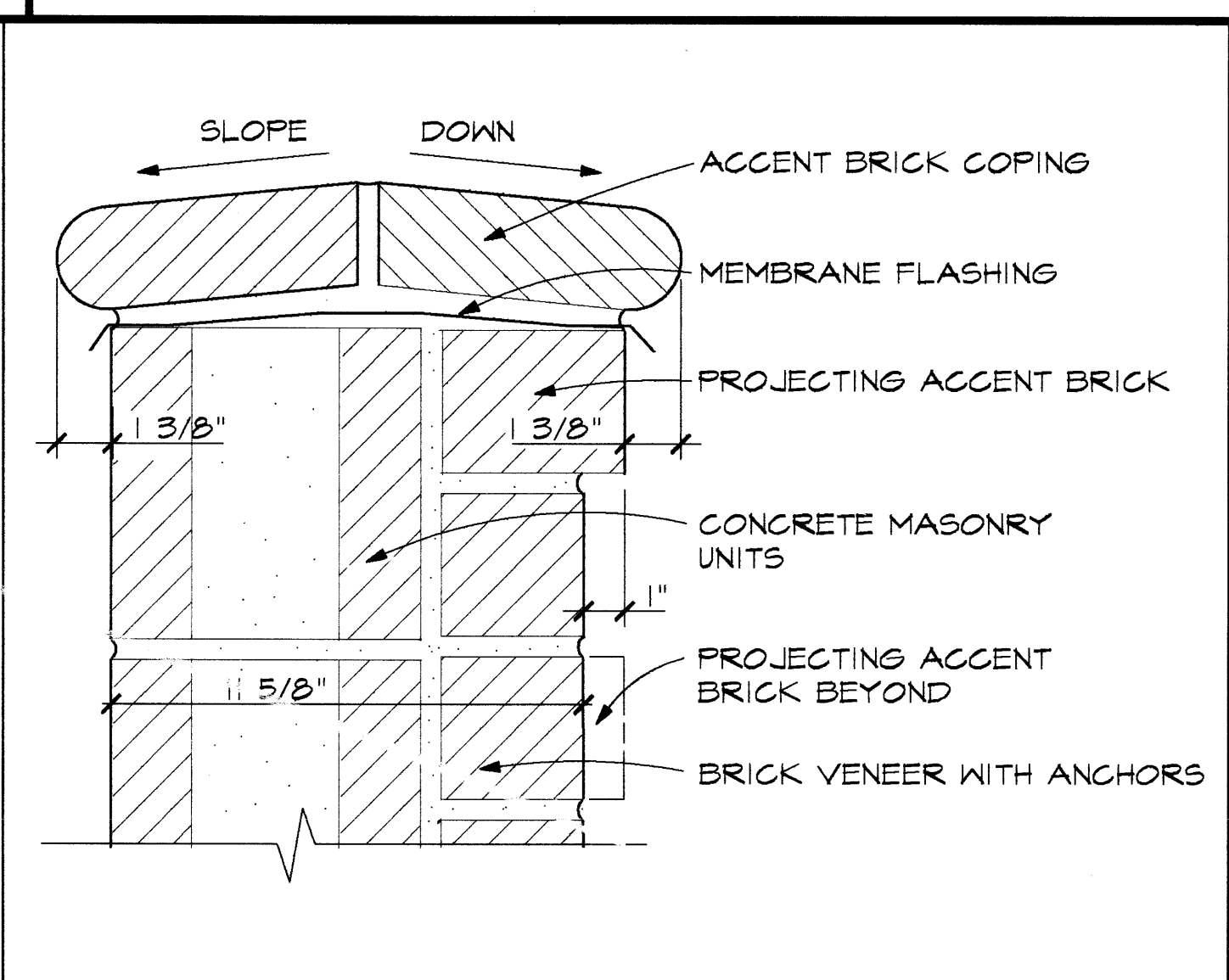
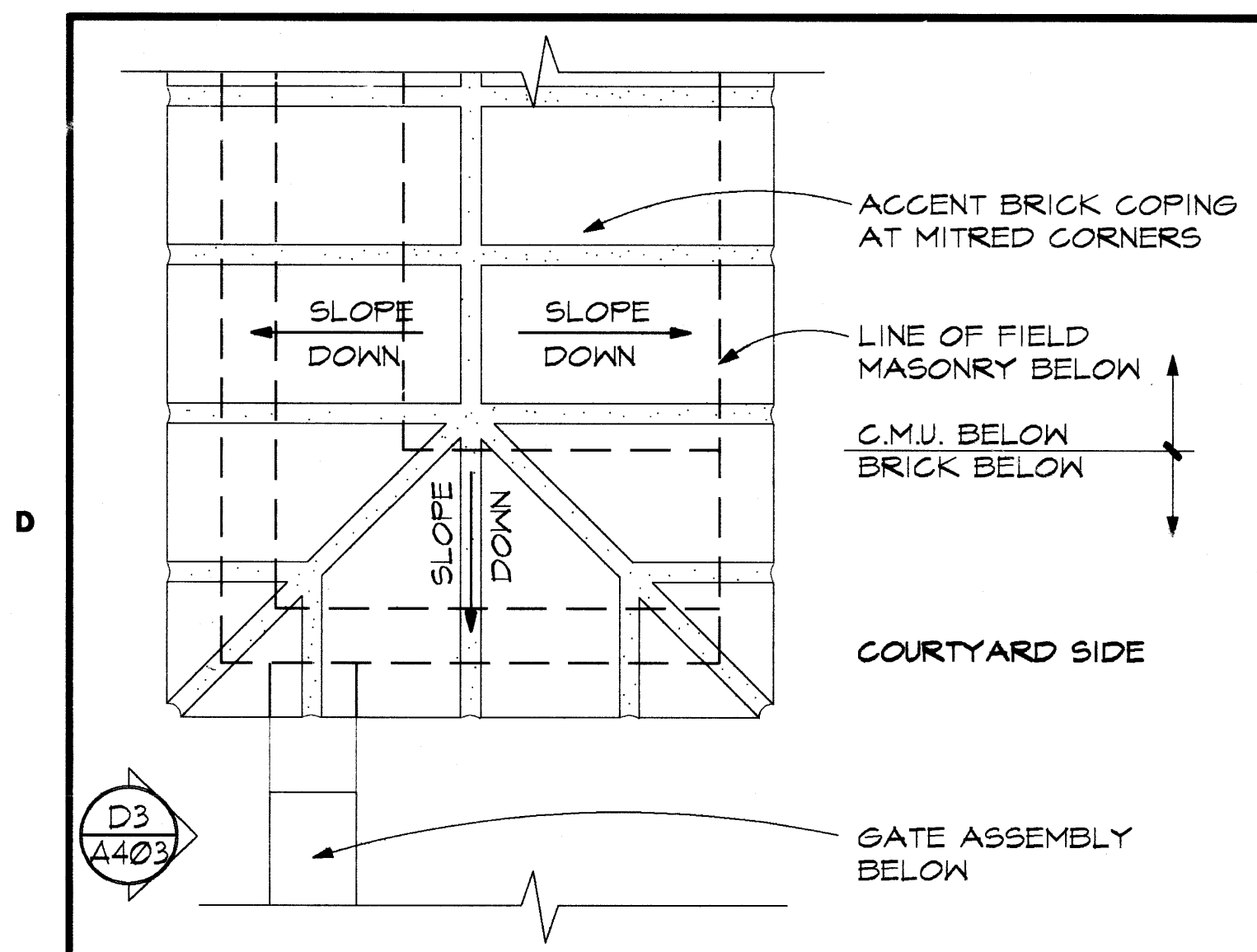
DETAILS

McCHORD AFB PN 33011 MC WASHINGTON

FILE NO. 2268/740-14-02 DATE 94 JUNE 16 A402

DESIGNER: ROCKOVSKY OK: B.K. SHEET 33

PLOT SCALE: FULL

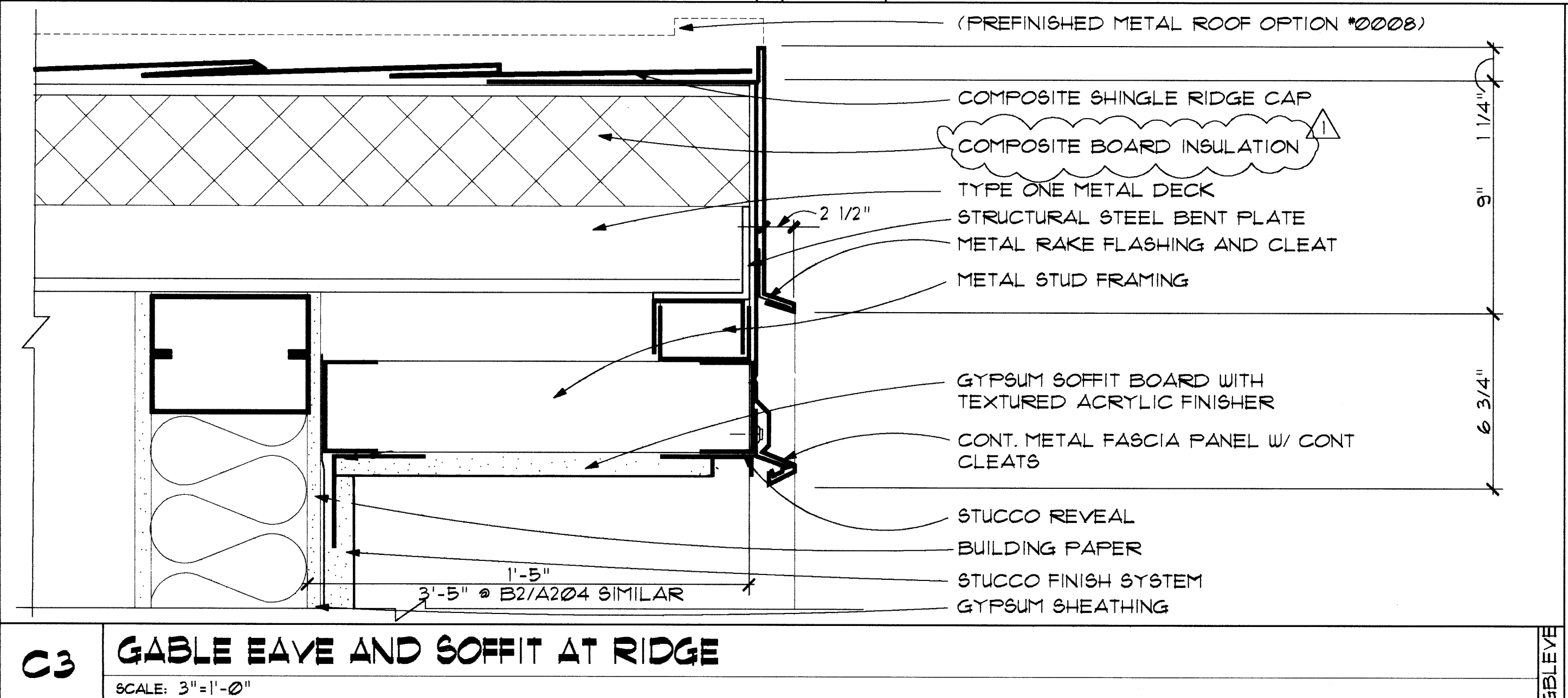
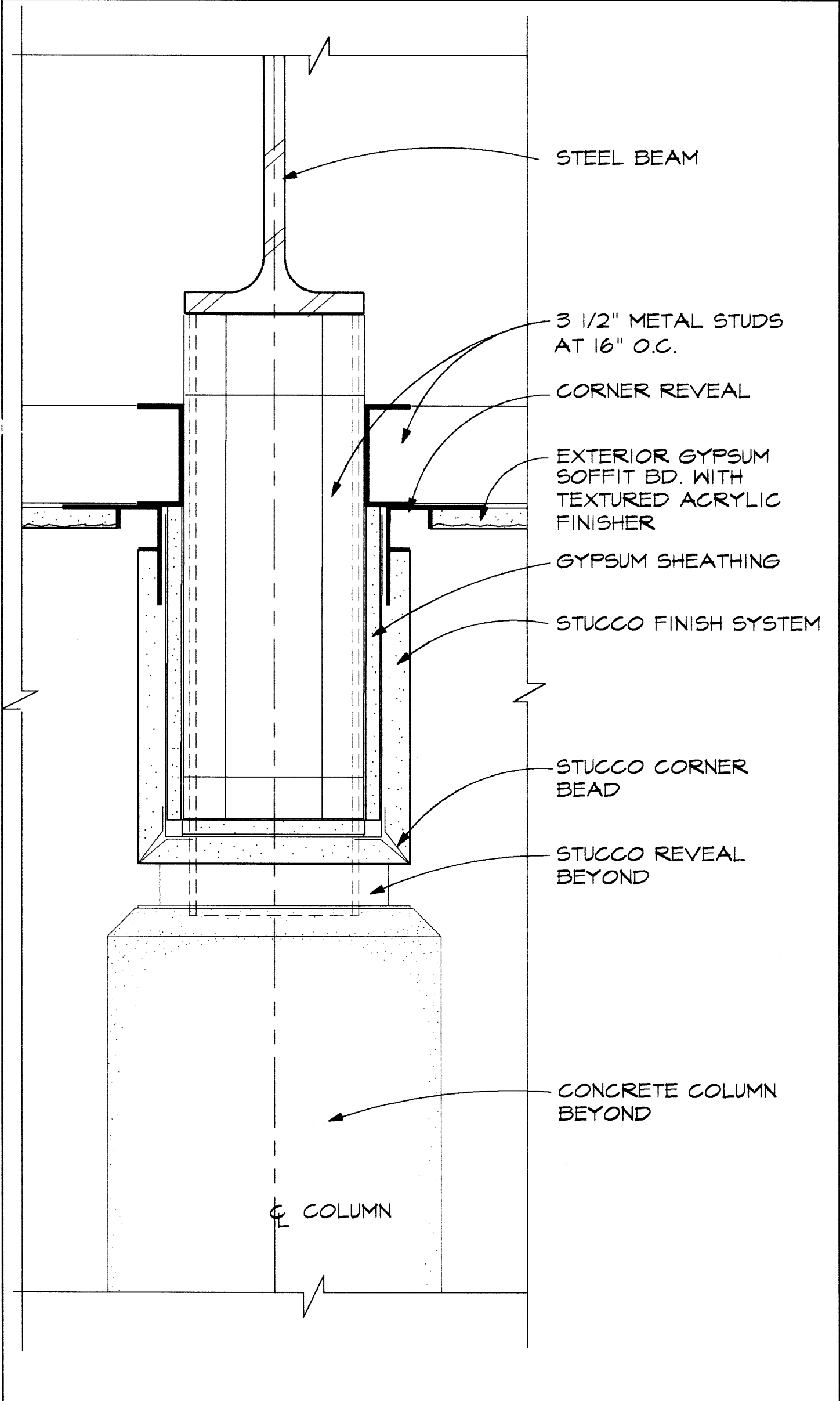
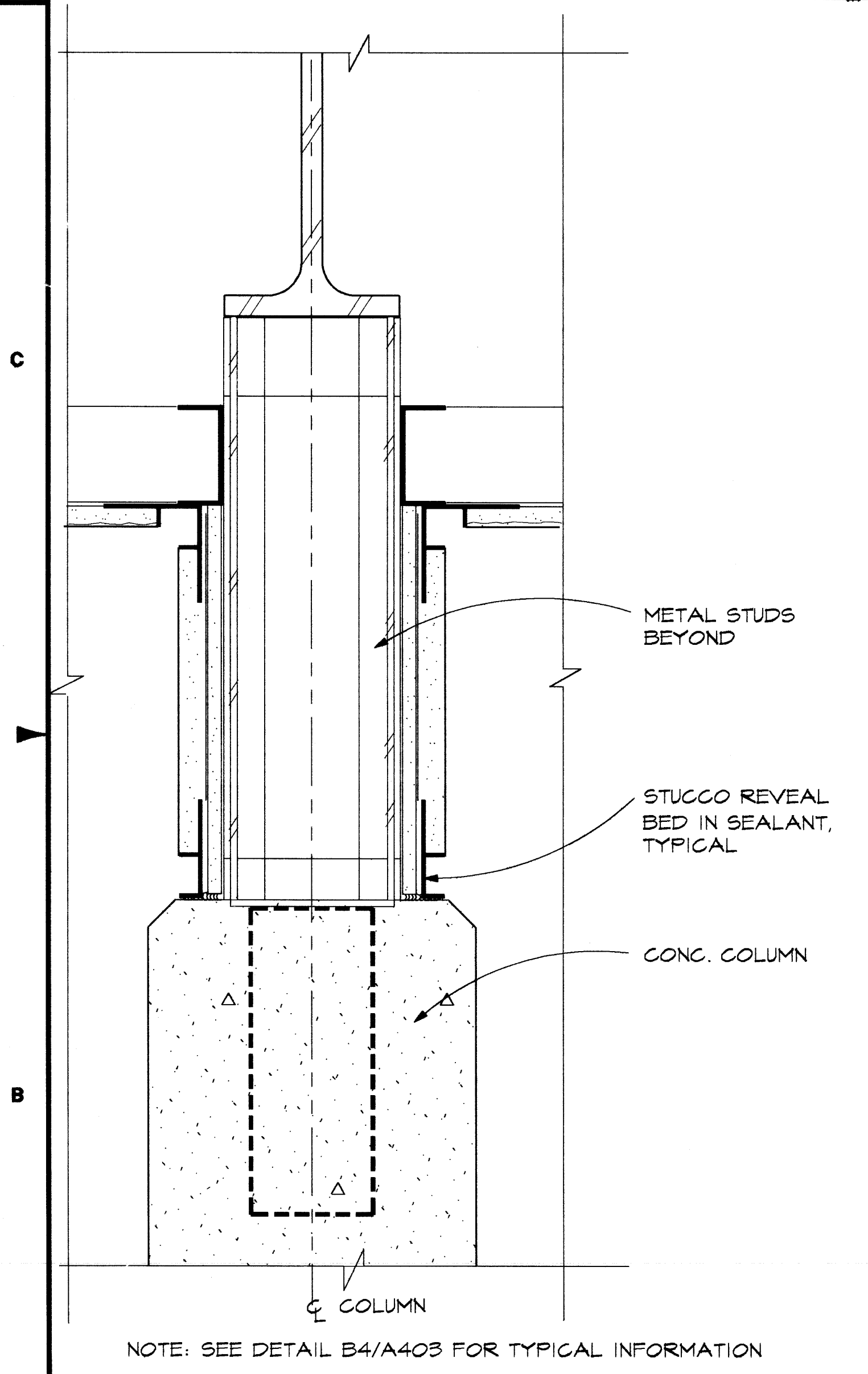


D5 BRICK CAP AT WALL
SCALE: 3"=1'-0"

D4 WALL AT MECHANICAL COURT
SCALE: 3"=1'-0"

D3 MECHANICAL GATE ELEVATION
SCALE: 1/2"=1'-0"

D2 AWNING ELEVATION
SCALE: 1/2"=1'-0"

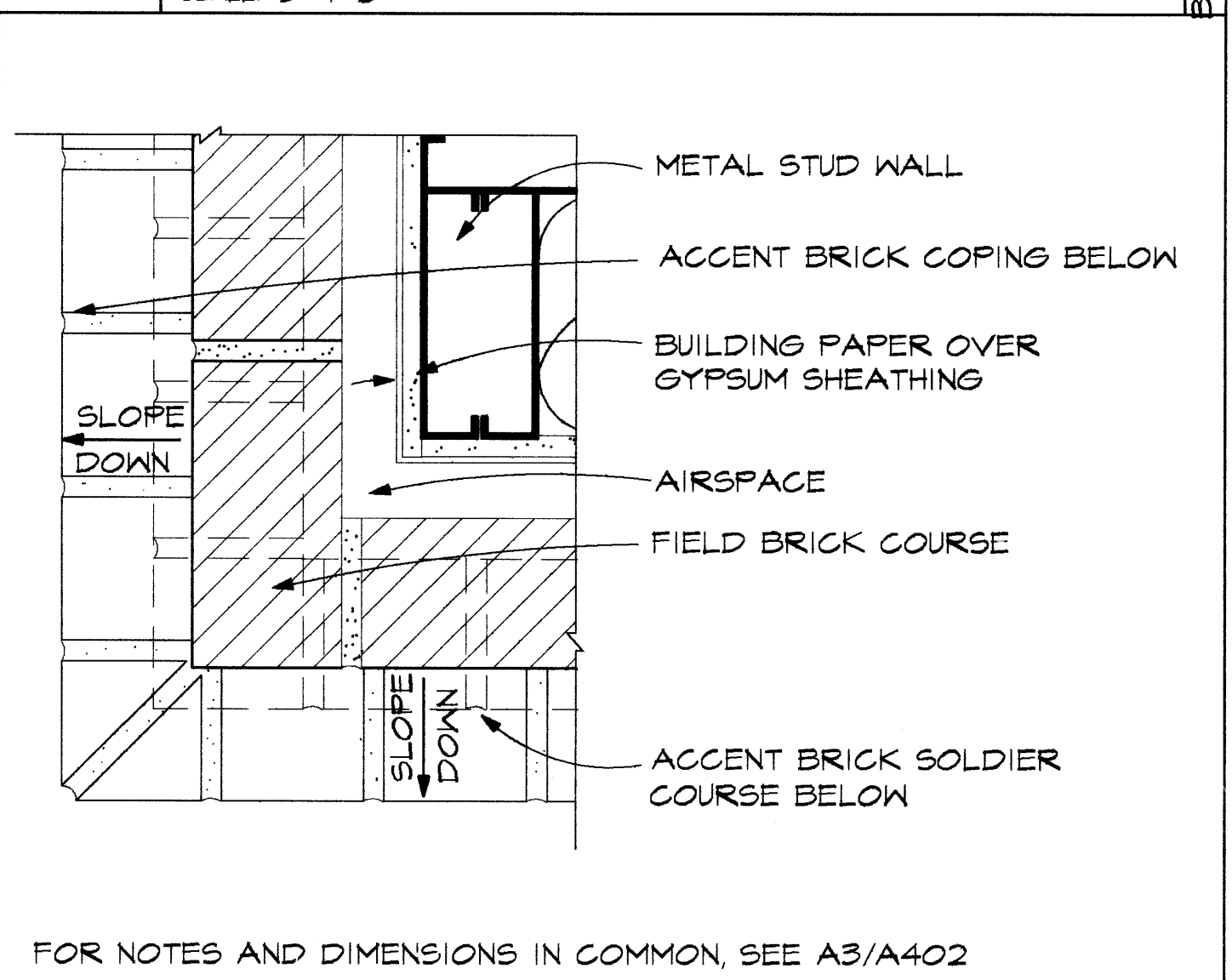
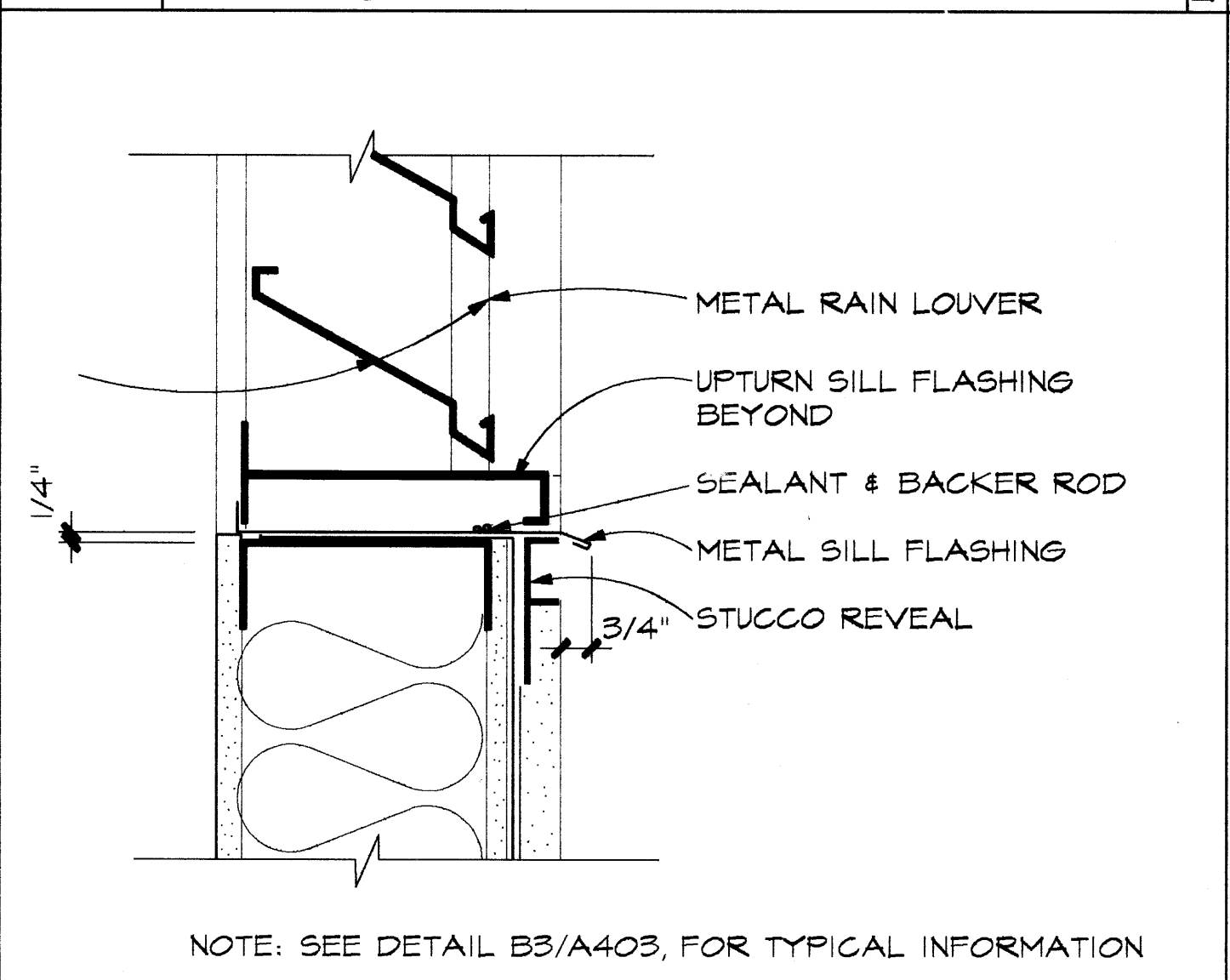
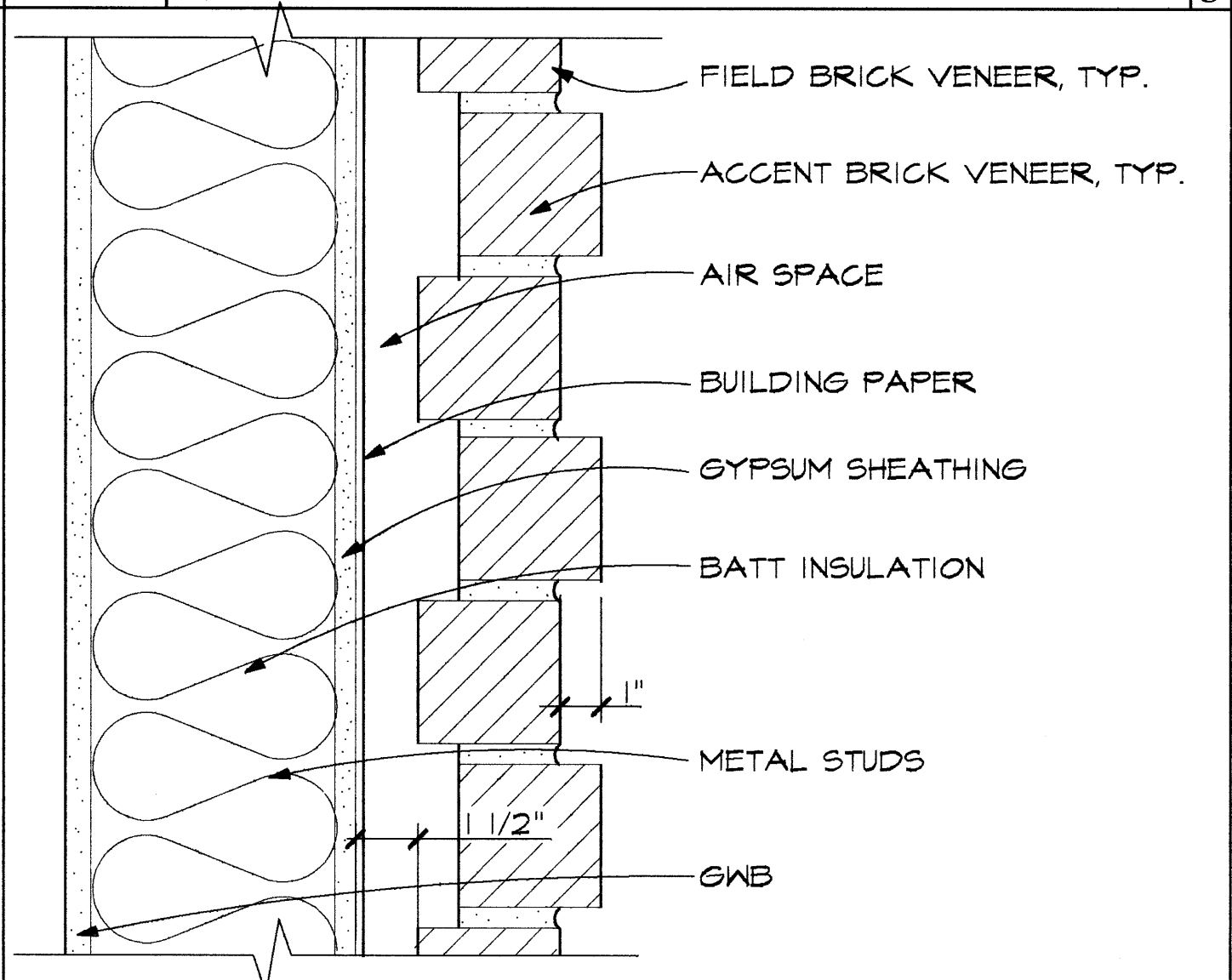
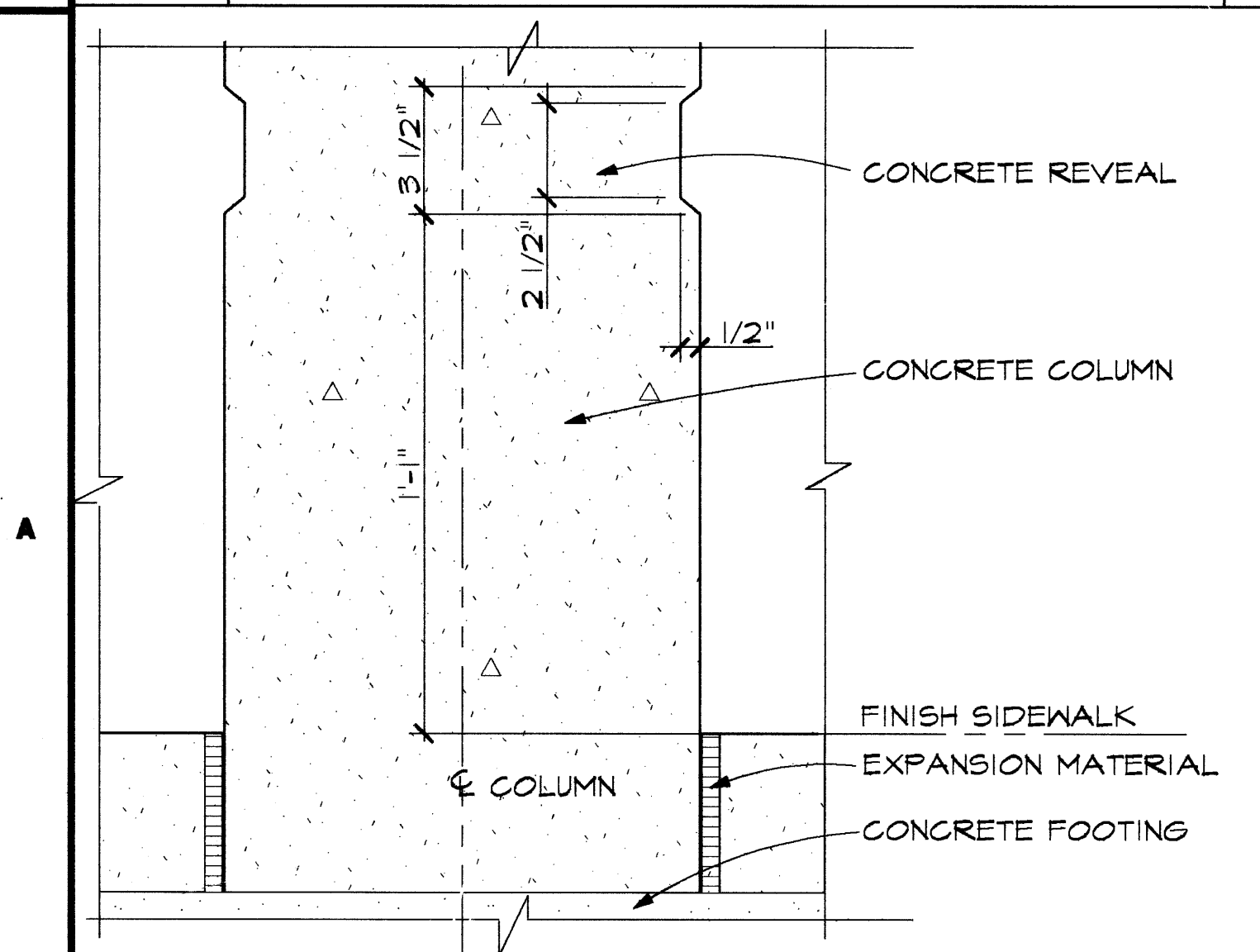


B5 COLUMN HEAD
SCALE: 3"=1'-0"

B4 COLUMN BEAM
SCALE: 3"=1'-0"

B3 LOUVER HEAD/JAMB
SCALE: 3"=1'-0"

B2 BRICK COPING CORNER-UPPER
SCALE: 3"=1'-0"



A5 COLUMN BASE AND REVEAL
SCALE: 3"=1'-0"

A4 SECTION AT OFFSET BRICK PATTERN
SCALE: 3"=1'-0"

A3 LOUVER SILL
SCALE: 3"=1'-0"

A2 BRICK COPING CORNER-LOWER
SCALE: 3"=1'-0"

REVISIONS				
SYMBOL	ZONE	DESCRIPTION	DATE	BY
△	C2,D2 SHEET	CHANGE IN WORDING TO COMPLY WITH SPECIFICATIONS REVISED AS-BUILT	94 JULY 5 26 FEB 19	RLR

AS - BUILT R 0001

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THEN THE DRAWING IS REDUCED SIZE

TSANG
ARCHITECTS

REGISTERED ARCHITECTS
ENGINEER DISTRICT, SEATTLE
CORPS OF ENGINEERS
SEATTLE, WASHINGTON

McCHORD CHILD DEVELOPMENT CENTER

DETAILS

McCHORD AFB PN 33011 MC WASHINGTON

FILE NO. 226S/740-14-02 DATE: 94 JUNE 16 PLATE: A403

DESIGN: ROCHKOVSKY CHK: ROCHKOVSKY SHEET: 34 OF 108